

Peritoneum and Peritoneal cavity

Department of Anatomy

The abdomen (3)

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Lecture points

- Definition of Peritoneum.
- Peritoneal reflections
- Peritoneal folds of the anterior abdominal wall
- Omentum:
- Lesser Omentum
- Epiploic foramen
- Greater Omentum
- Greater Omentum related ligaments
- Peritoneal compartments
- Peritoneal Ligaments
- Peritoneal Folds
- Peritoneal cavity
- Lesser Sac (Omental Bursa)
- Greater Sac (Omental Bursa)

Peritoneum

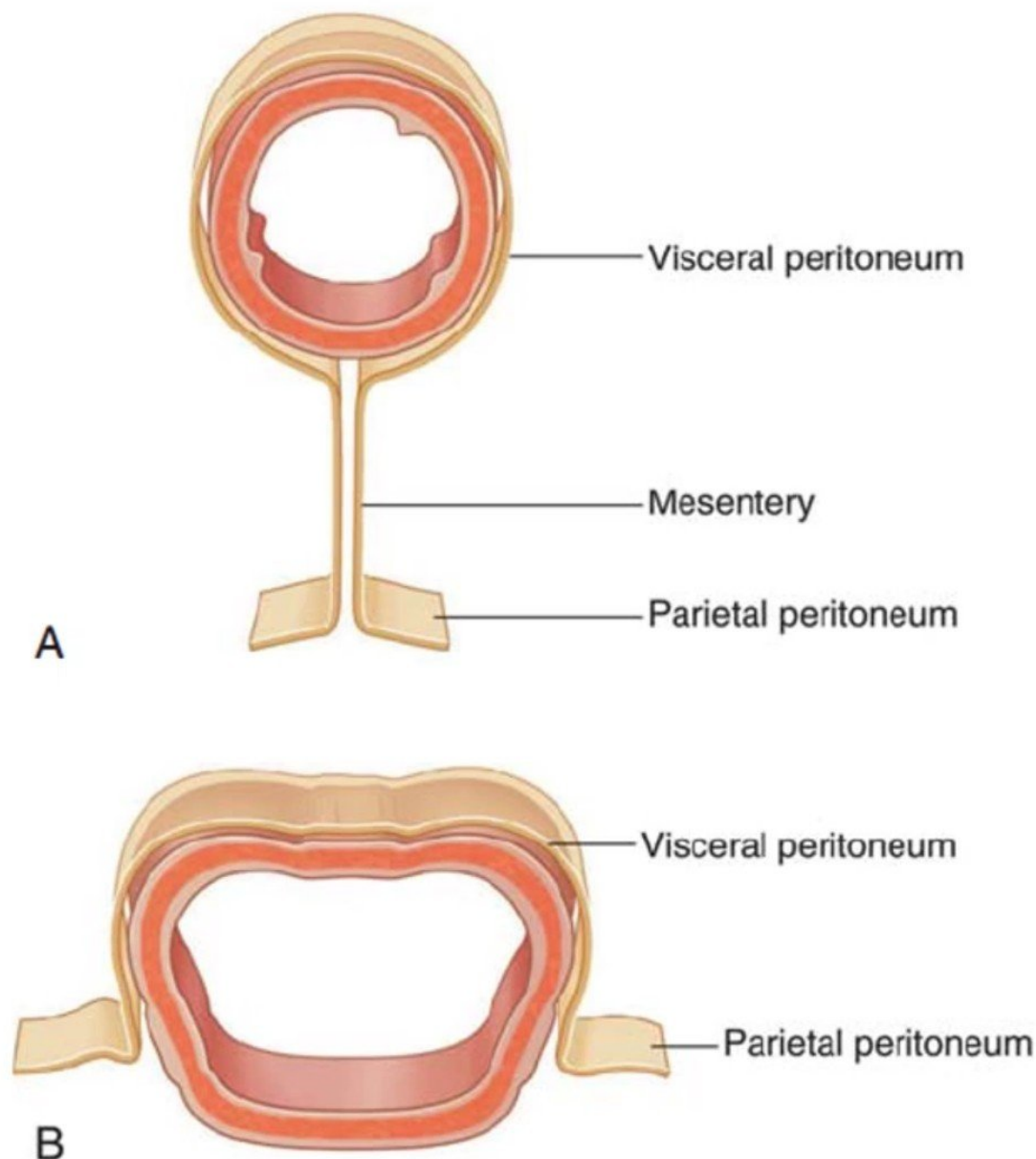
- It is a **serous membrane** lined by mesothelial cells.
- It consists of the **parietal peritoneum** and the **visceral peritoneum**.

A) Parietal Peritoneum.

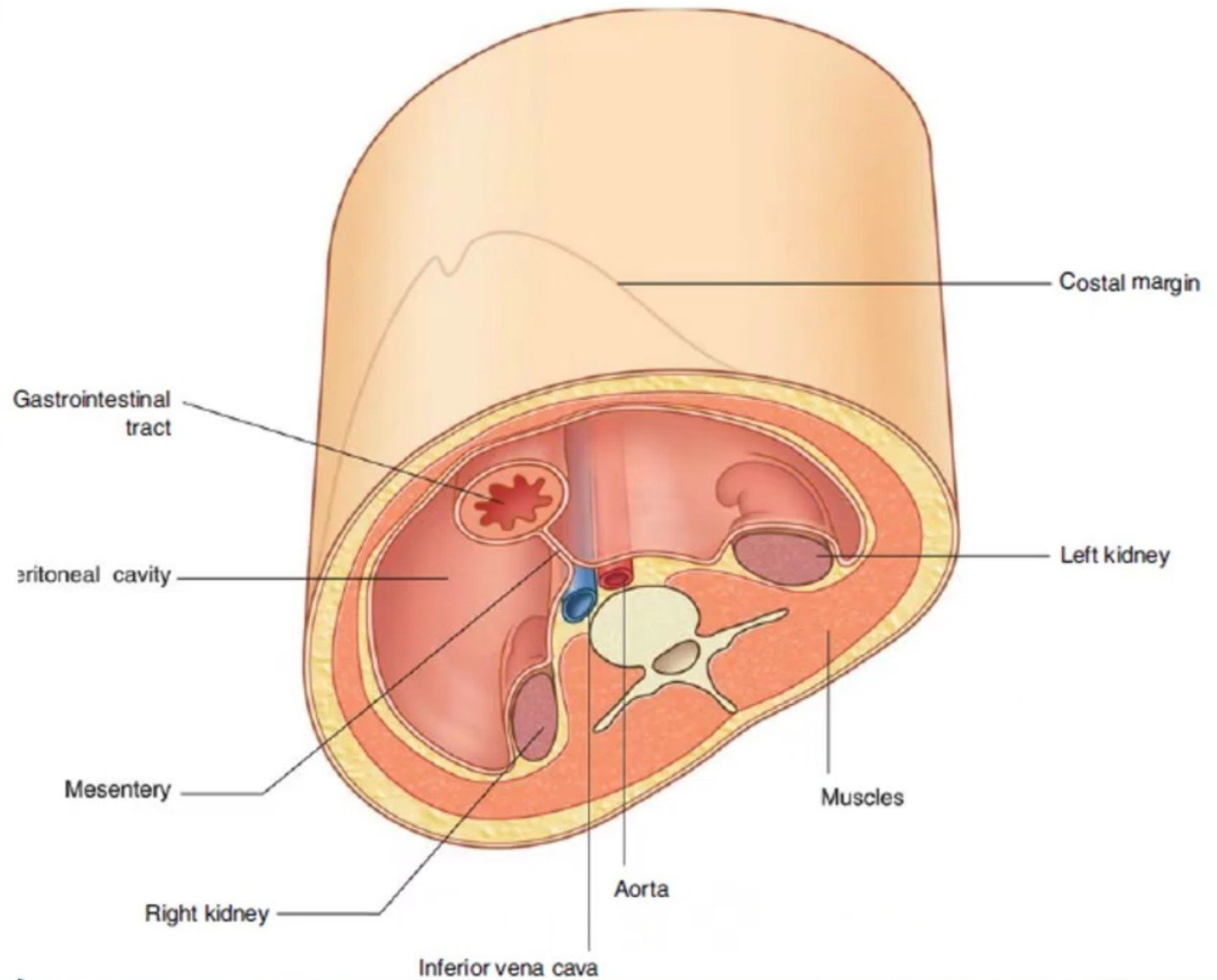
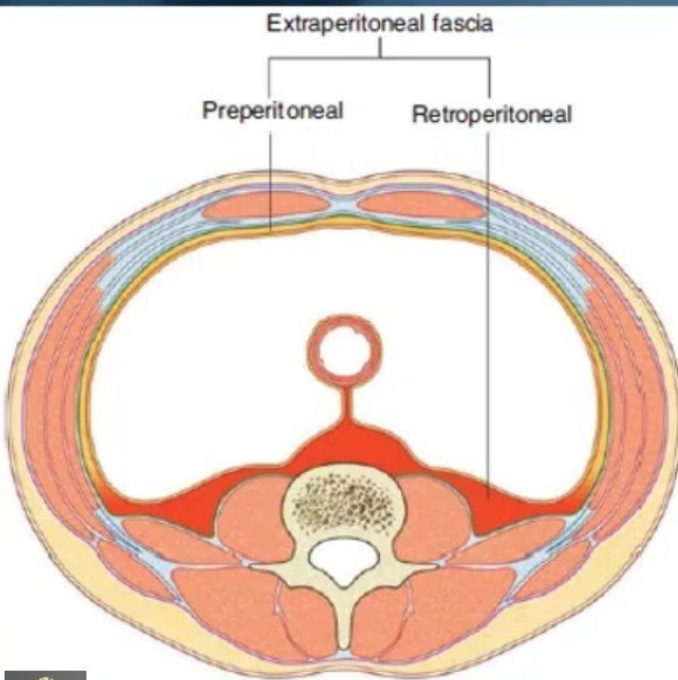
- It lines the abdominal and pelvic walls and the inferior surface of the diaphragm.
- It is innervated by **somatic nerves** such as **the phrenic, lower intercostal, subcostal, iliohypogastric, and ilioinguinal nerves**.

B) Visceral Peritoneum.

- It **covers the viscera**.
- It is **innervated by visceral nerves**.
- It is **insensitive to pain**.

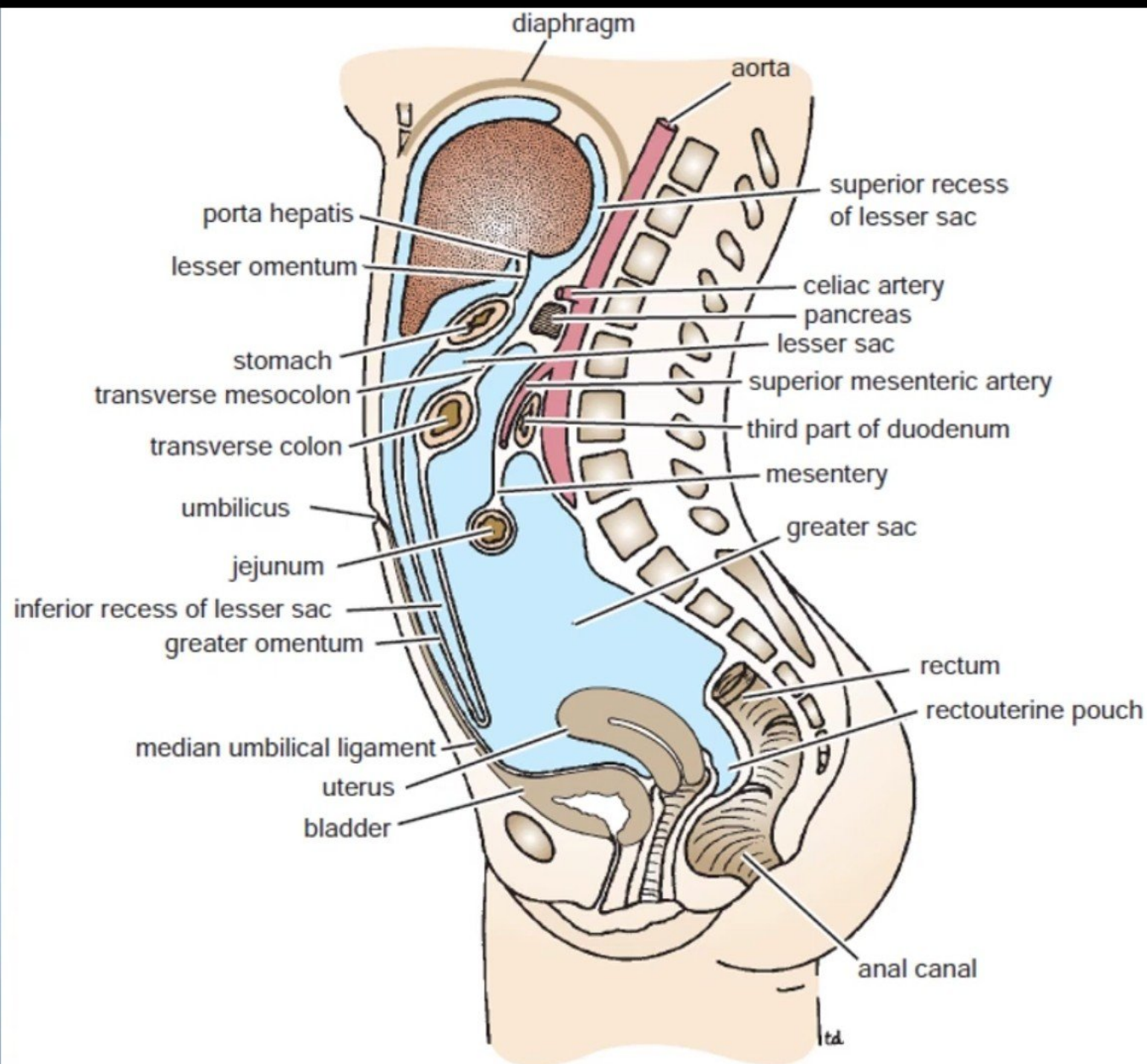


Peritoneum



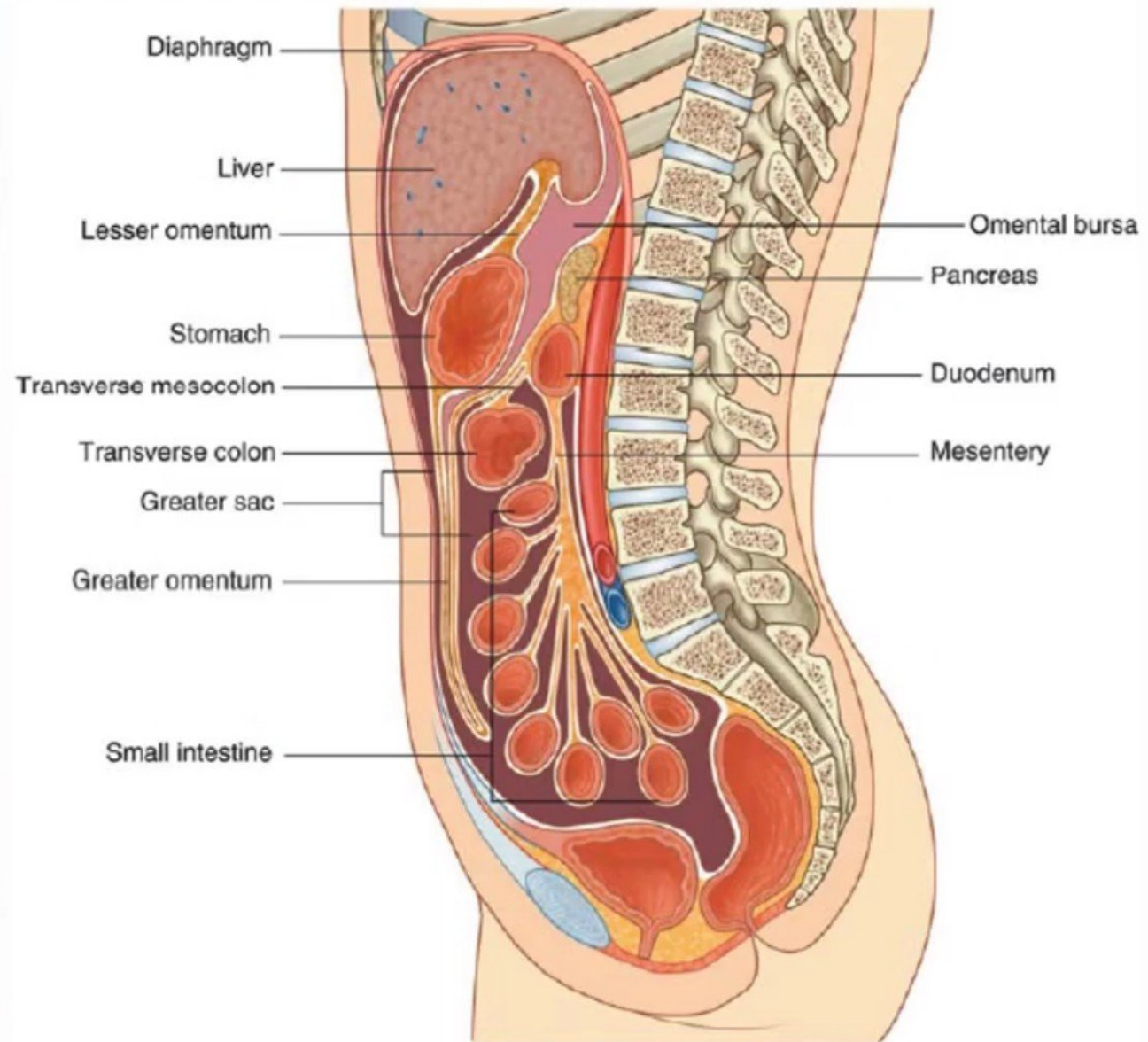
Intraperitoneal and Retroperitoneal Relationships

- ✓ The terms **intraperitoneal** and **retroperitoneal** are used to describe **the relationship of various organs to their peritoneal covering**.
- ✓ An organ is said to be **intraperitoneal** when it is **almost totally covered with visceral peritoneum**. The stomach, jejunum, ileum, and spleen are good examples of intraperitoneal organs.
- ✓ **Retroperitoneal** organs lie **behind the peritoneum** and **are only partially covered with visceral peritoneum**. The pancreas and the ascending and descending parts of the colon are examples of retroperitoneal organs.
- ✓ No organ, however, is actually within the **peritoneal cavity**.
- ✓ An intraperitoneal organ, such as the stomach, appears to be **surrounded by the peritoneal cavity**, but it is covered with visceral peritoneum and is attached to other organs by omentum.



Peritoneum

- The **parietal peritoneum**.
- The **visceral peritoneum**.
- **Various folds or reflexions of peritoneum** connect viscera to the abdominal walls or to one another. Some of these are properly called **folds**, but others are called: **mesentery, omentum** or **ligament**.



Peritoneal reflections

The mesentery:

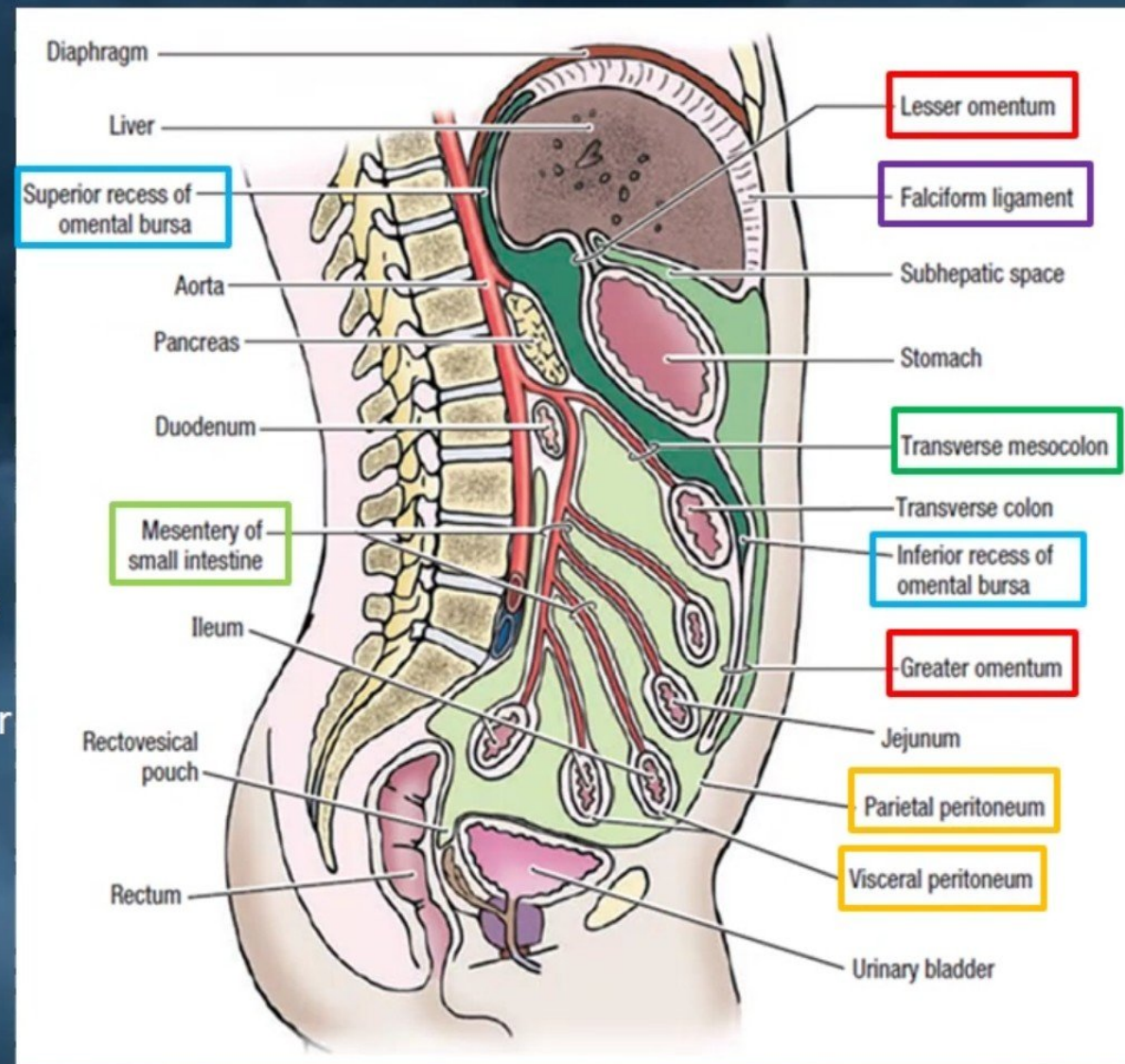
- The double fold **supporting the small intestine**.
- when the **mesenteries supporting the large intestine** named:
 - The **transverse colon** called the **transverse mesocolon**.
 - The **sigmoid colon** called **sigmoid mesocolon**.
 - The **appendix** called **mesoappendix**.

The omentum:

- The **lesser omentum** connects the stomach to the liver,
- The **greater omentum** hangs down from the lower border of the stomach.

The ligament:

- The **various ligaments** associated with the liver, stomach and spleen are simply peritoneal folds attached to them, and the broad ligament stretches out on either side of the uterus.



Peritoneal reflections

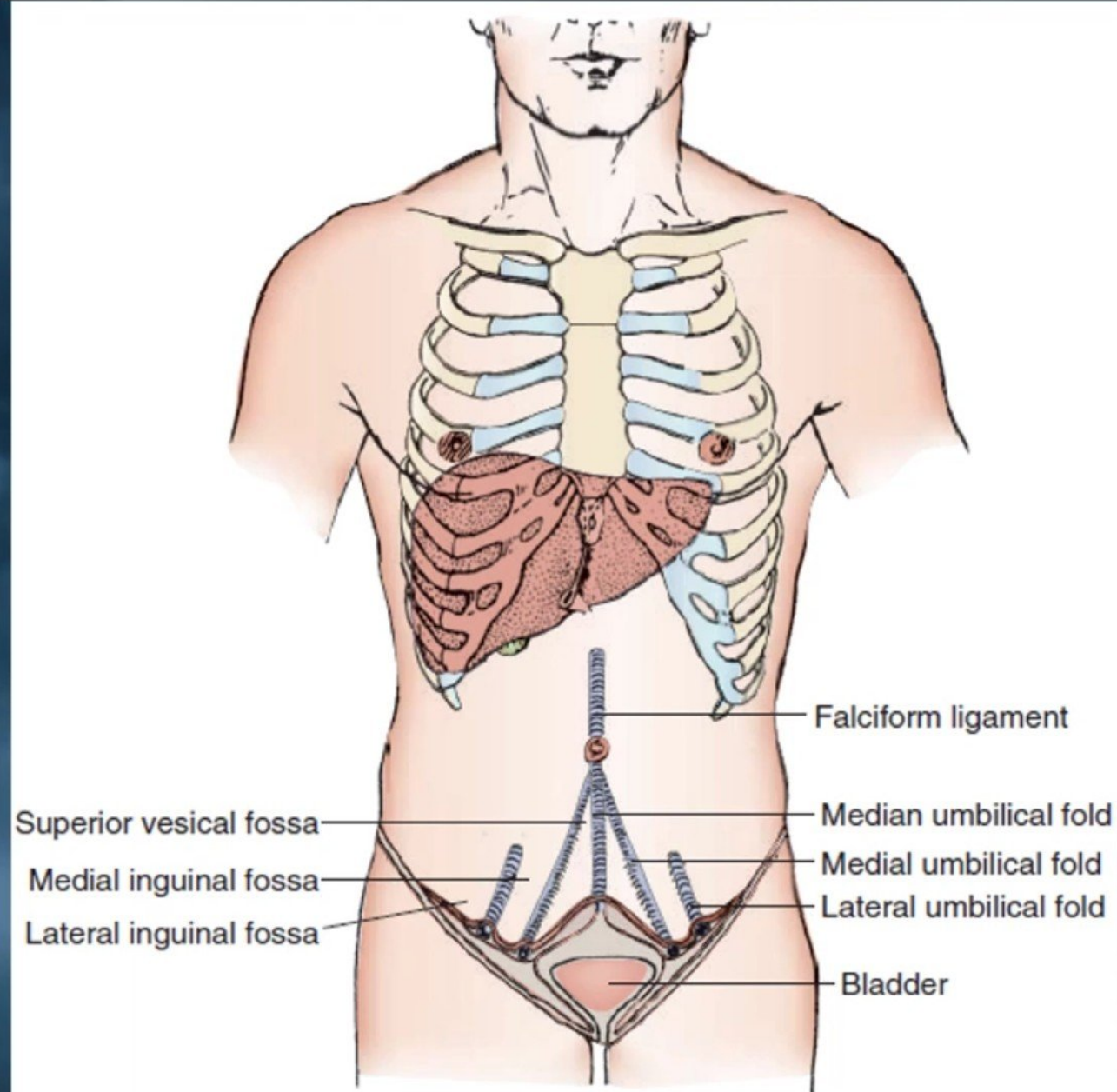
Term	Definition
Peritoneal ligament	Double layer of peritoneum that connects an organ with another organ or to the abdominal wall.
Mesentery	Double layer of peritoneum that occurs as a result of the invagination of the peritoneum by an organ and constitutes a continuity of the visceral and parietal peritoneum.
Omentum	Double – layered extension of peritoneum passing from the stomach and proximal part of the duodenum to adjacent organs. ▪ The greater omentum extends from the greater curvature of the stomach. ▪ The lesser omentum from the lesser curvature of the stomach.
Bare area	Every organ must have an area, the bare area, that is not covered with visceral peritoneum to allow the entrance and exit of neurovascular structures. Bare areas are formed in relation to the attachments of mesenteries, omenta, and ligaments. Named bare area eg bare area of liver are especially extensive in area.

Peritoneal folds of the anterior abdominal wall

- On the posterior surface of the anterior abdominal wall the peritoneum is raised into **six folds**, **one above** and **five below** the umbilicus.

1) The falciform ligament (one):

- It consists of two adherent layers of peritoneum.
- It connecting **the anterior surface of the liver** to **the supra-umbilical part of the anterior abdominal wall**, (just to the right of the midline), and to **the inferior surface of the diaphragm**.
- Its concave, inferior margin, which **contains the ligamentum teres** (the obliterated remains of the left umbilical vein), deviates to the right and is attached to the notch for this ligament on the inferior border of the liver.



Peritoneal folds of the anterior abdominal wall

- Below the umbilicus there is a central fold with a pair on either side.

2) The median umbilical fold (one):

- It is found centrally.
- It is containing **the median umbilical ligament** (the remains of the urachus).

3) The medial umbilical fold (two):

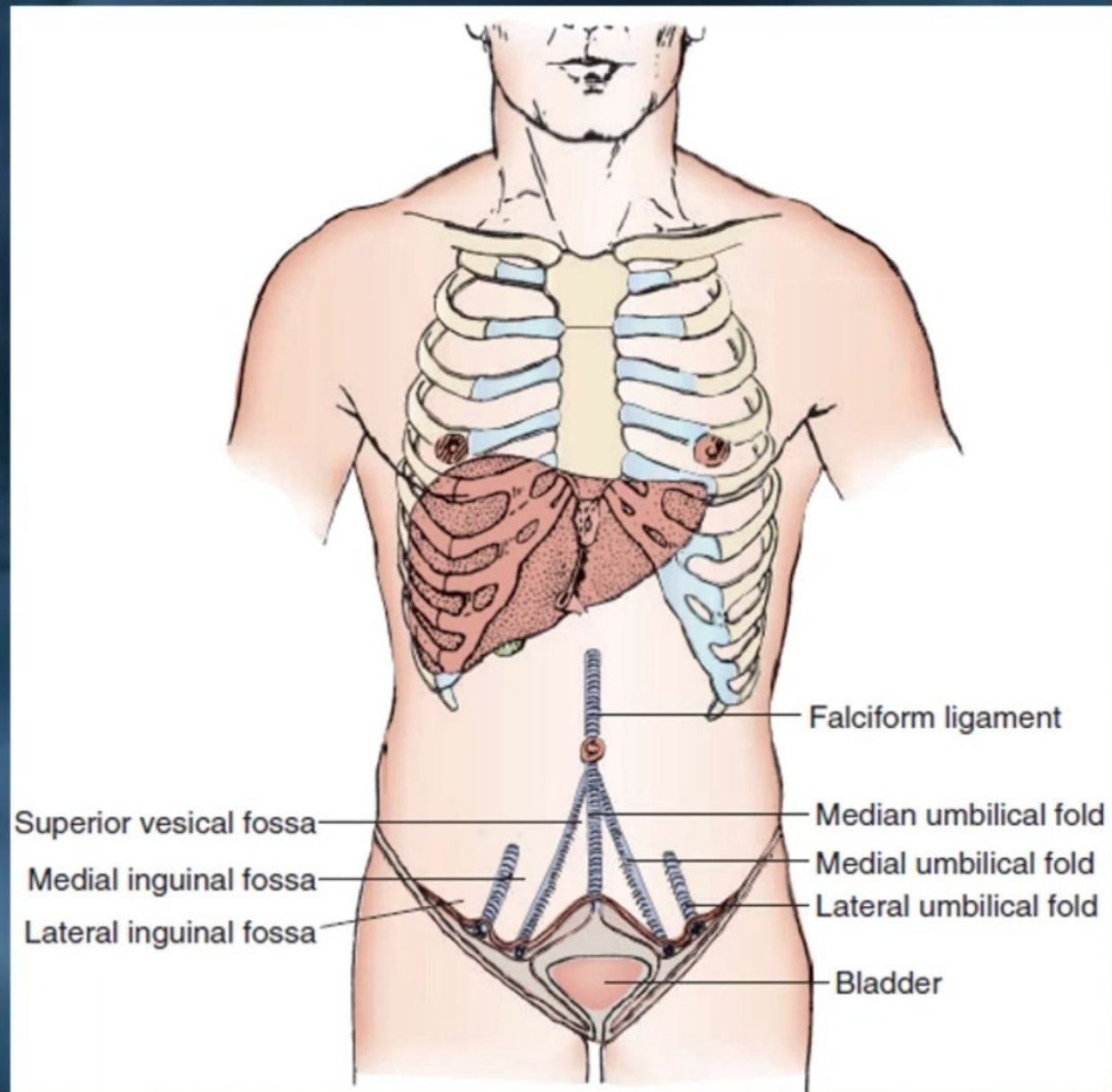
- It is found on each side.
- It is running as far as the umbilicus.
- It is containing the medial umbilical ligament (**the obliterated remains of the umbilical artery**).

4) The lateral umbilical fold (two):

- It is found farther laterally
- It is containing **the inferior epigastric vessels**,
- which enter the rectus sheath by passing across the arcuate line;

Note:

although called umbilical folds, this lateral pair do not reach as far as the umbilicus.



Omentum

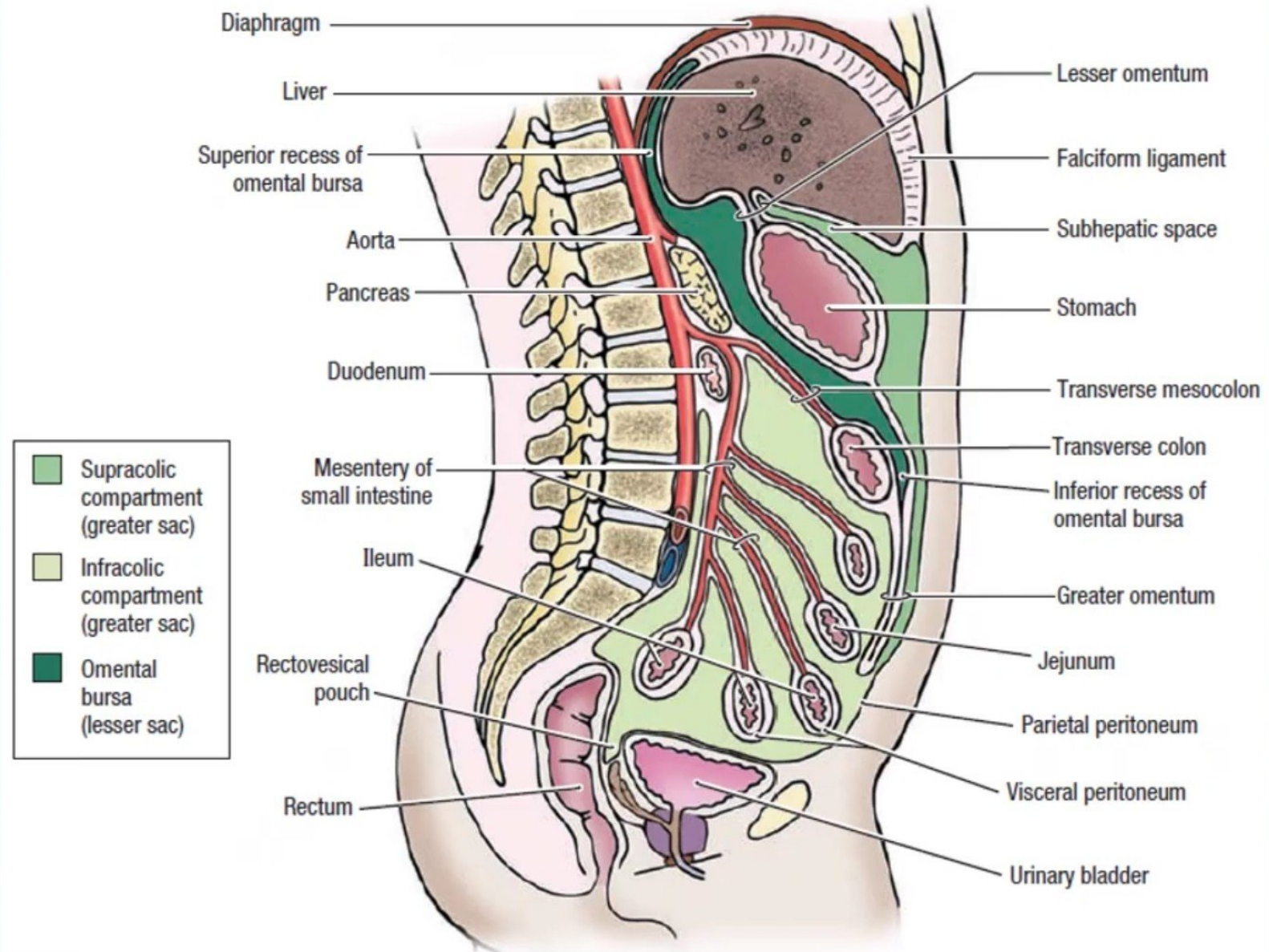
- It support the viscera and provide pathways for associated neurovascular structures.

Omentum:

- It is a fold of peritoneum extending from the stomach to adjacent abdominal organs.

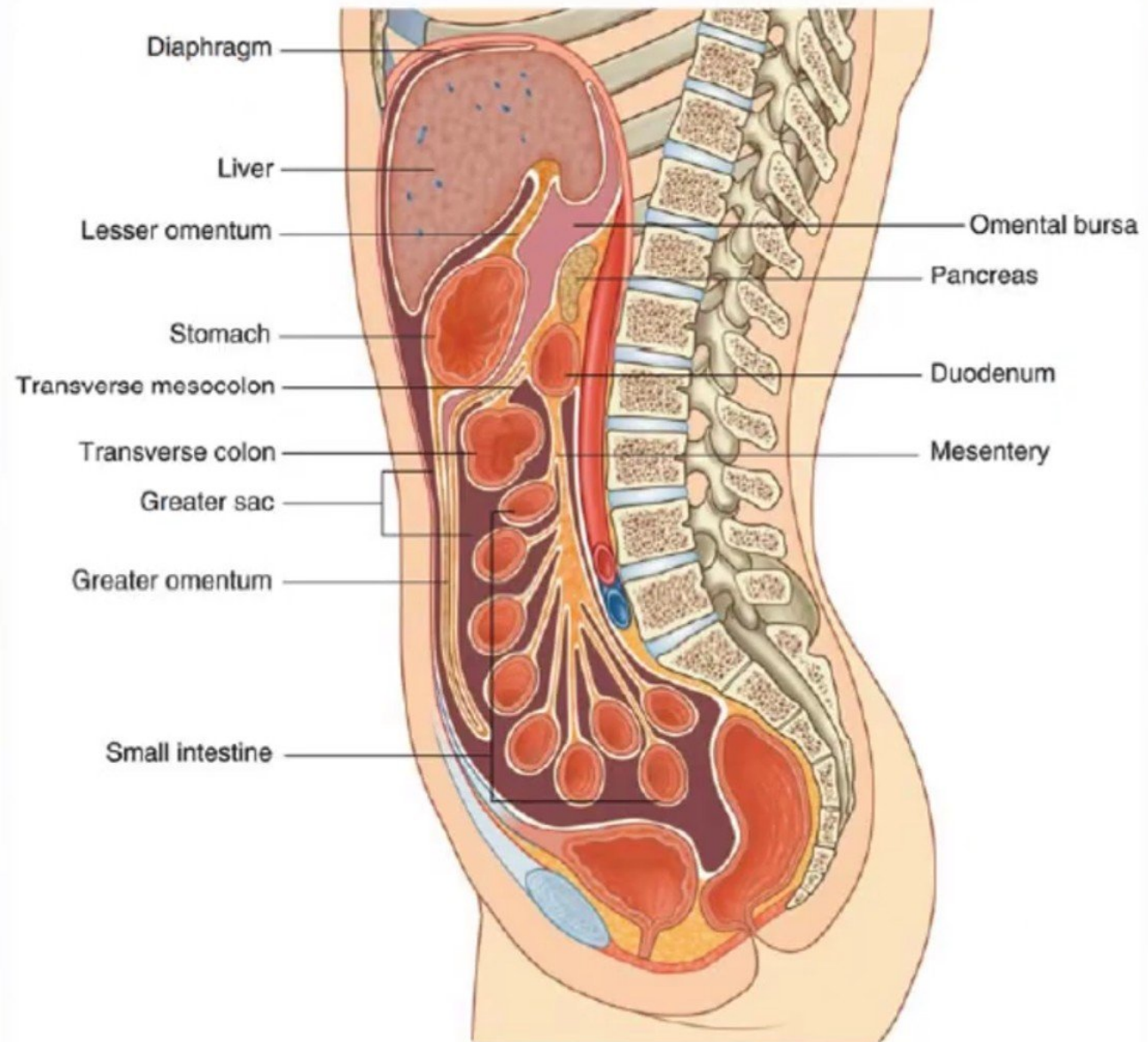
1. Lesser Omentum.

2. greeter Omentum.



Lesser Omentum

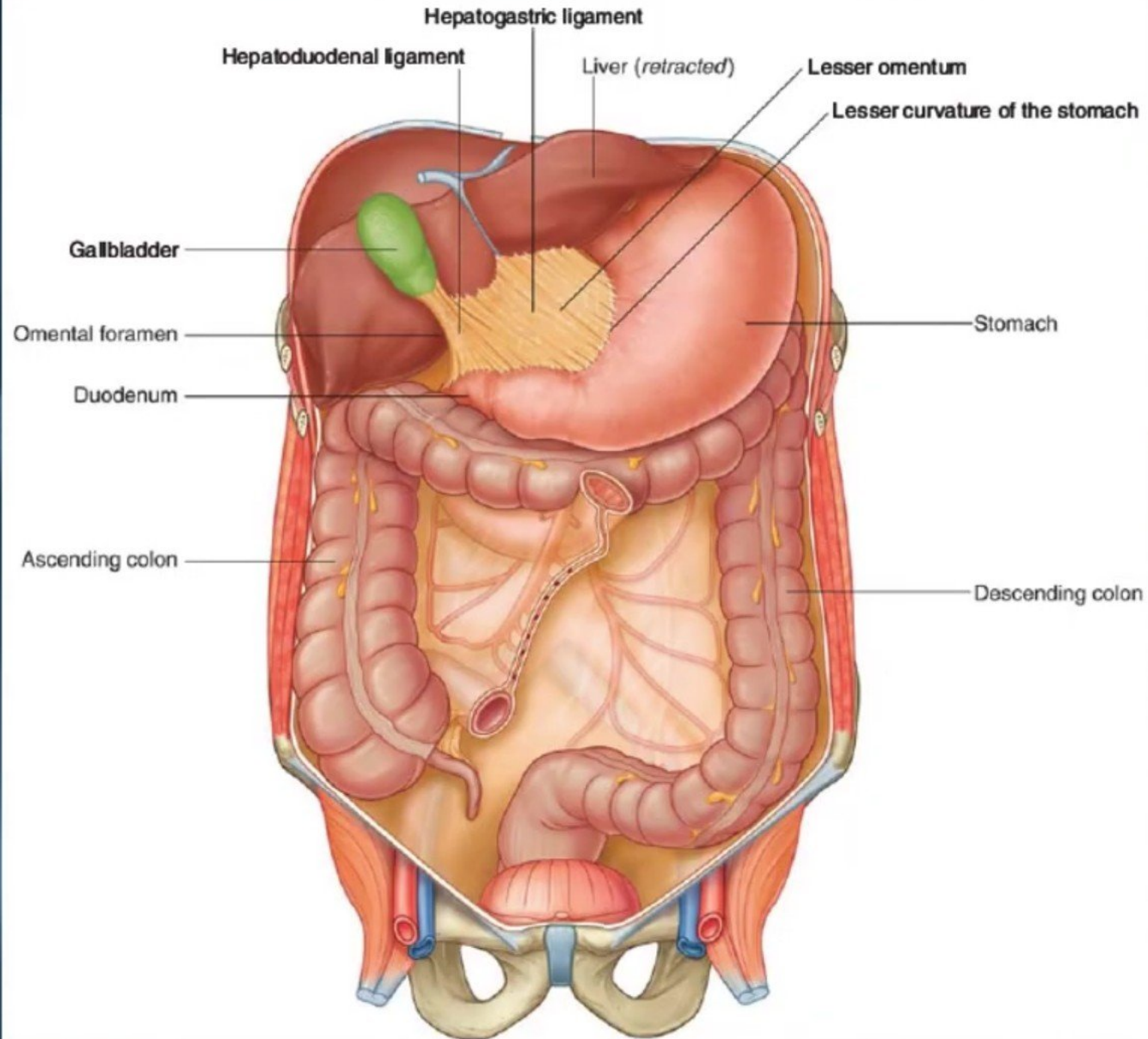
- It is a **double layer of peritoneum** extending from **the porta hepatis of the liver** to **the lesser curvature of the stomach** and the **beginning of the duodenum**.
- It consists of the **hepatogastric** and **hepatoduodenal ligaments**.
- It forms the **anterior wall of the lesser sac** of the peritoneal cavity.
- Transmits **the left and right gastric vessels**, which **run between its two layers along the lesser curvature**.
- Has a right free margin that contains the **proper hepatic artery, bile duct, and portal vein**.



Epiploic foramen

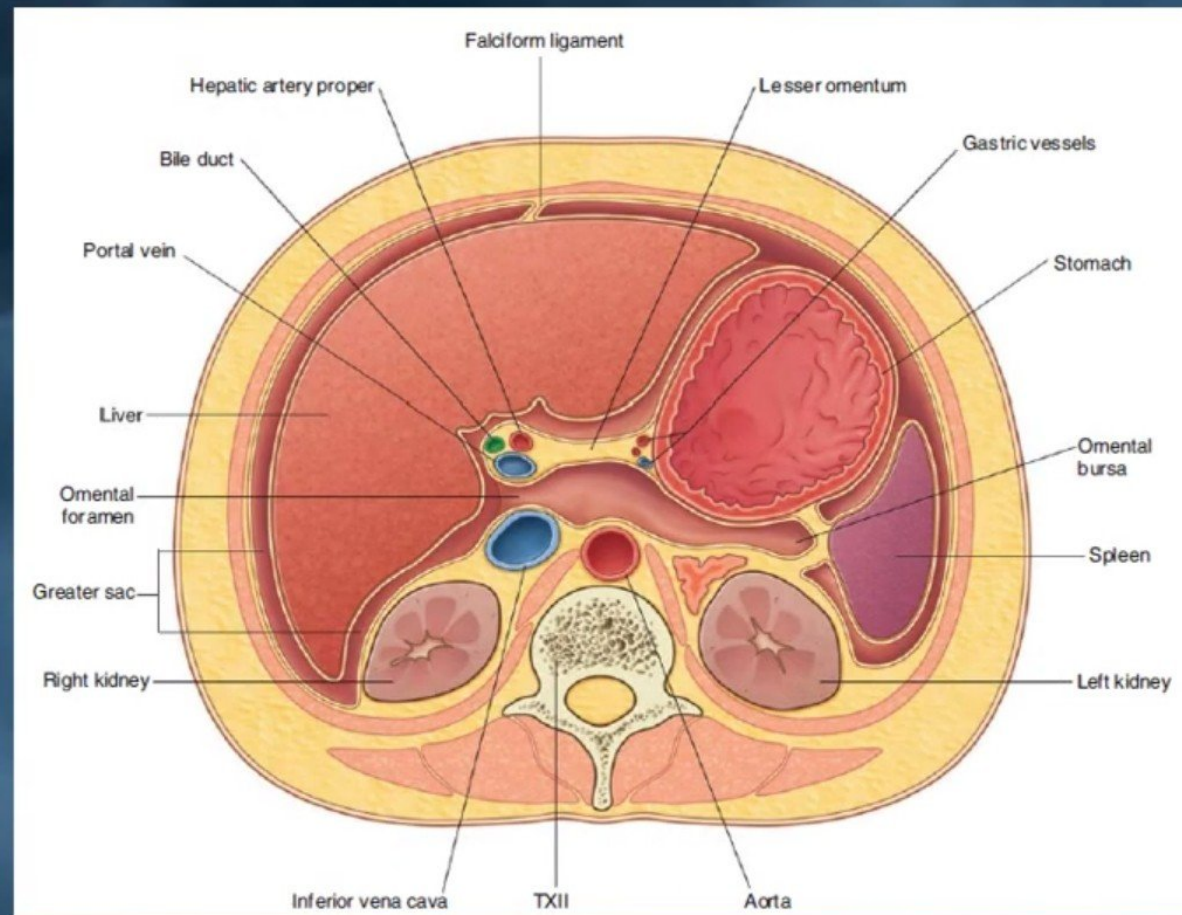
The epiploic foramen (of Winslow) (omental foramen):

- The adieus to the lesser sac
- It is foramen between the **first part of duodenum (in the lower boundary)** and the caudate process of the **liver (in upper boundary)**.
- It has a **right free margin**, where the **anterior** and **posterior layers of peritoneum** become **continuous**.
- The liver attachment is L-shaped to the *fissure for the ligamentum venosum* and the **porta hepatis**. (hepatic artery proper, bile ducts and portal vein).
- This fold contain porta hepatis forms the **anterior boundary** of **epiploic foramen**.



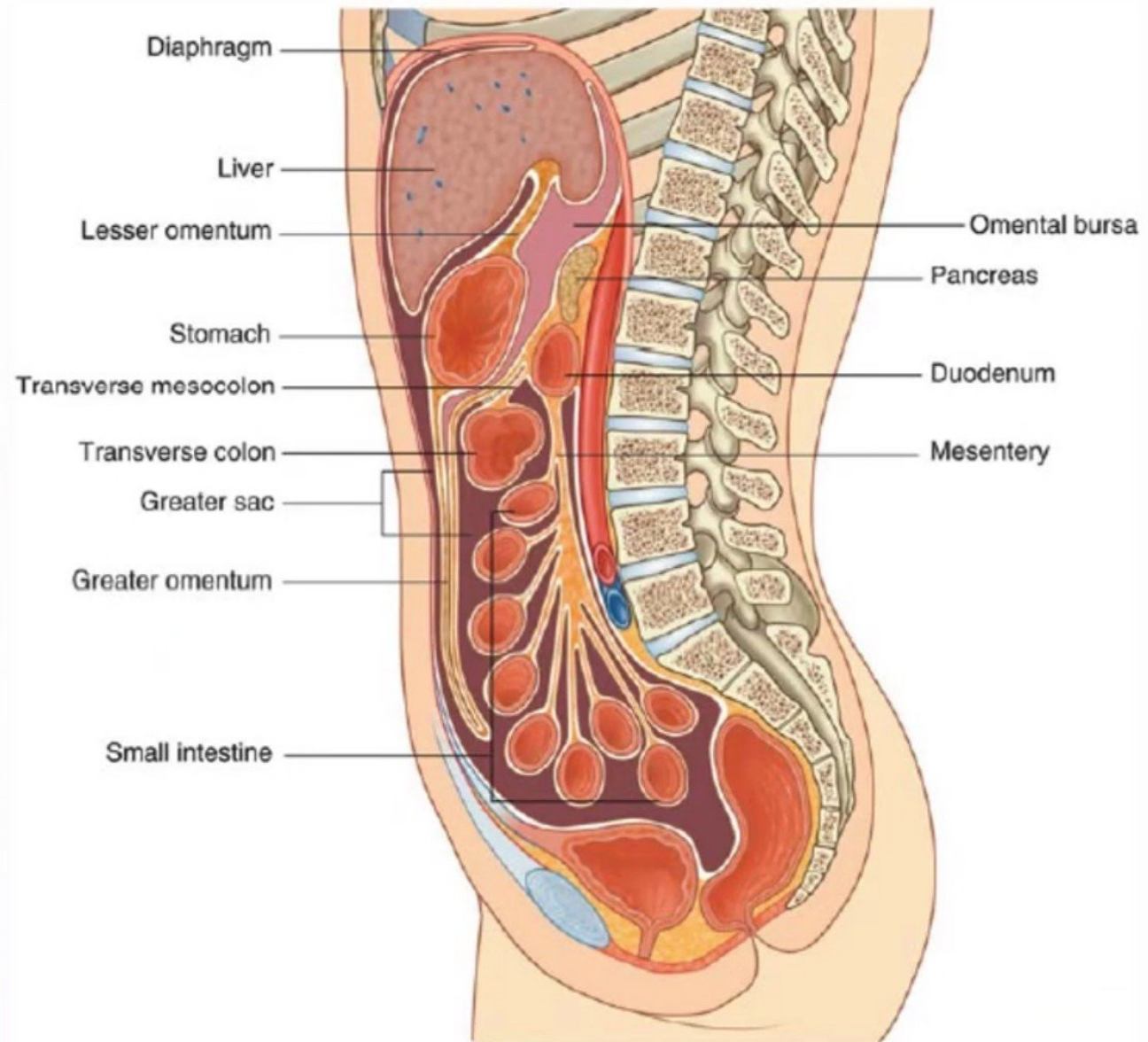
Epiploic or Omental (Winslow's) Foramen

- It is a natural opening between **the lesser** and **greater sacs**.
- It is bounded:
 - **Superiorly by** peritoneum on the **caudate lobe of the liver**.
 - **Inferiorly by** peritoneum on the **first part of the duodenum**.
 - **Anteriorly by** the **free edge of the lesser omentum**.
 - **Posteriorly by** peritoneum covering the **IVC**.



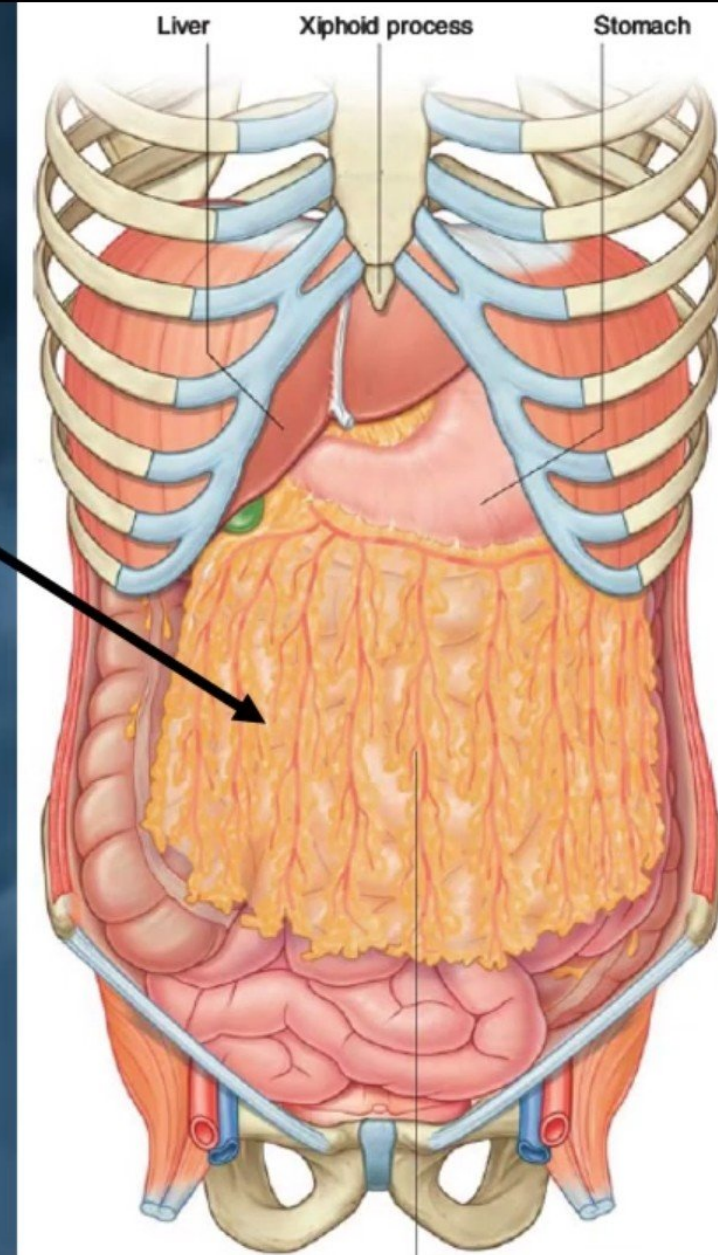
Greater Omentum

- It is **derived** from the **embryonic dorsal mesentery**.
- It **hangs down like an apron** from the **greater curvature of the stomach**, covering the **transverse colon** and other **abdominal viscera**.
- The **lesser sac** may be **accessed** through the **gastrocolic omentum**.
- Other routes of surgical access to the **lesser sac** are **through the lesser omentum** and **through the transverse mesocolon**.



Greater Omentum

- It transmits **the right and left gastroepiploic vessels** along the **greater curvature**.
- The greater omentum is **a double sheet of peritoneum, folded on itself to form four layers**.
- **The two layers of the lesser omentum become the greater omentum.**
- **The four layers of the greater omentum** below the transverse colon **fuse** with each other to form an integral structure.
- This contains **adipose tissue** of variable amount, and **macrophages**.
- The part of the greater omentum between the **stomach** and the **transverse colon** is often referred to as the **gastrocolic omentum**.
- **The right and left gastroepiploic vessels run between the layers of the gastrocolic omentum**, close to the greater curvature of the stomach.



Greater Omentum related ligaments

A) Gastrolenal (Gastrosplenic) Ligament:

- ✓ It extends from **the left portion of the greater curvature** of the stomach to **the hilus of the spleen**.
- ✓ It contains **the short gastric** and **left gastroepiploic vessels**.

B) Lienorenal (Splenorenal) Ligament:

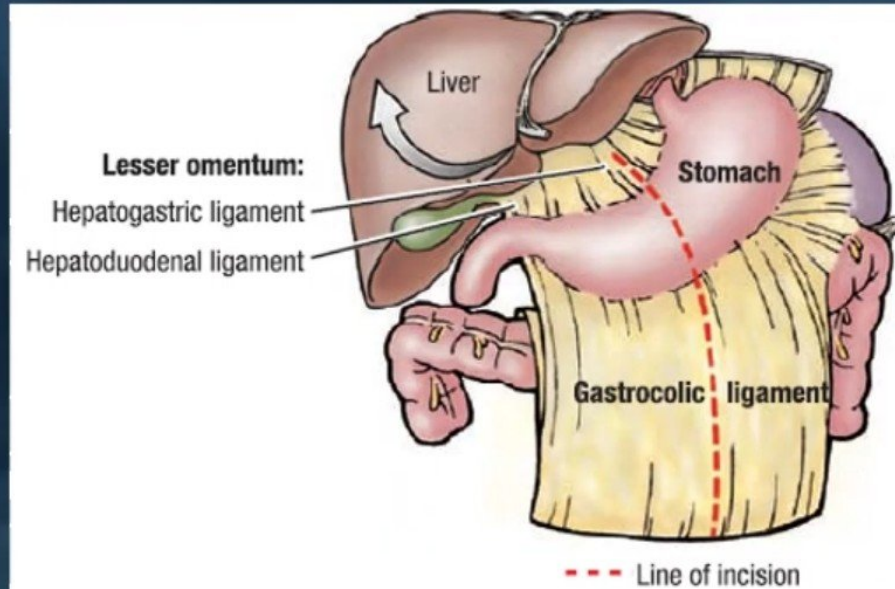
- ✓ It runs from **the hilus of the spleen** to **the left kidney**.
- ✓ It contains **the splenic vessels** and **the tail of the pancreas**.

C) Gastrophrenic Ligament:

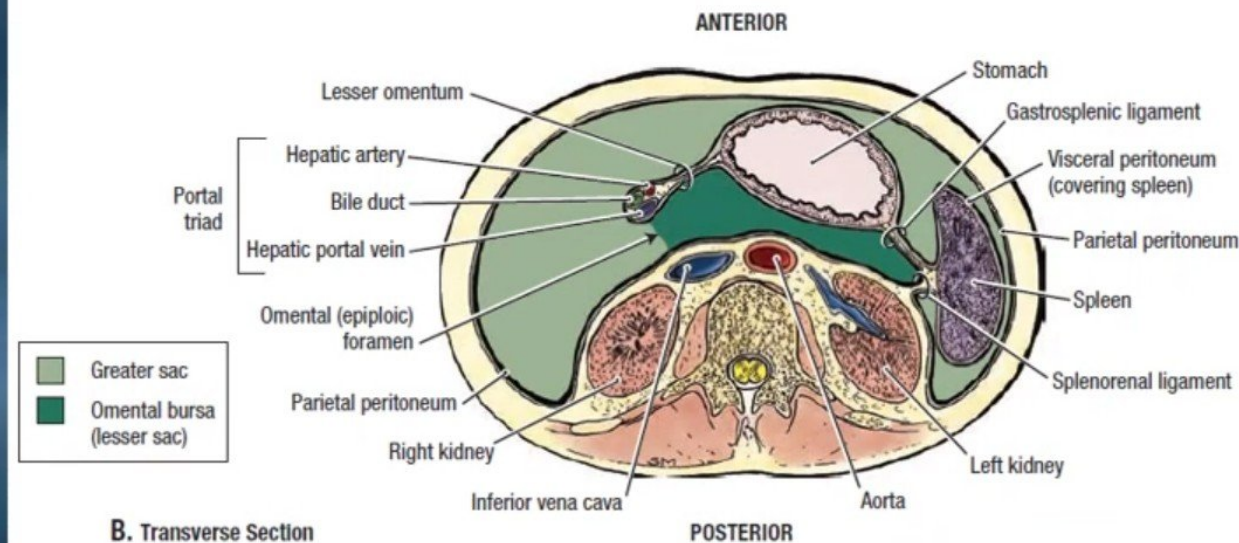
- ✓ It runs from the upper part of the greater curvature of the **stomach** to **the diaphragm**.

D) Gastrocolic Ligament:

- ✓ It runs from the greater curvature of **the stomach** to the **transverse colon**.



B. Anterior View



B. Transverse Section

Peritoneal compartments

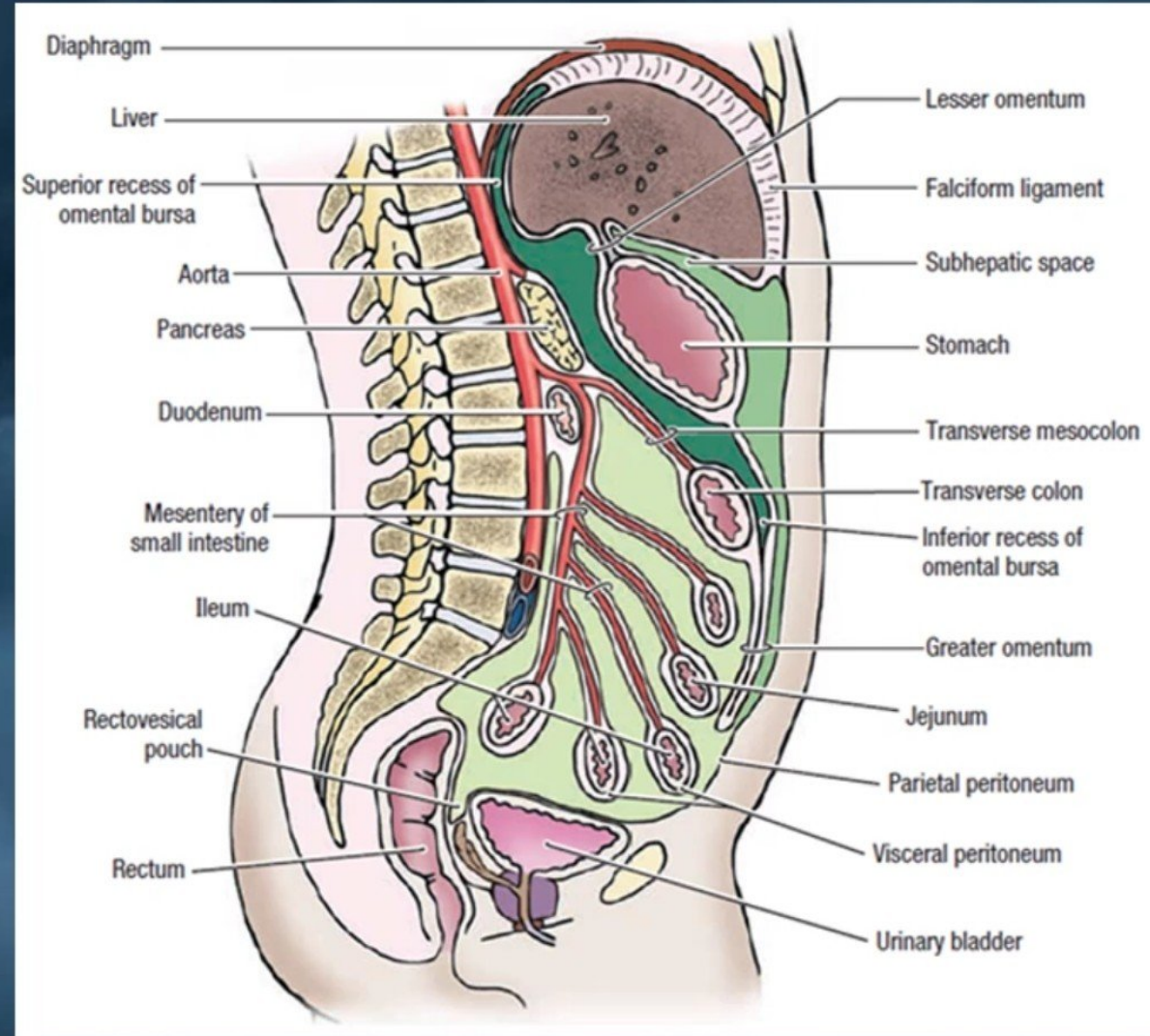
The peritoneal cavity is descriptively divided into compartments called:

✓ **Supracolic.**

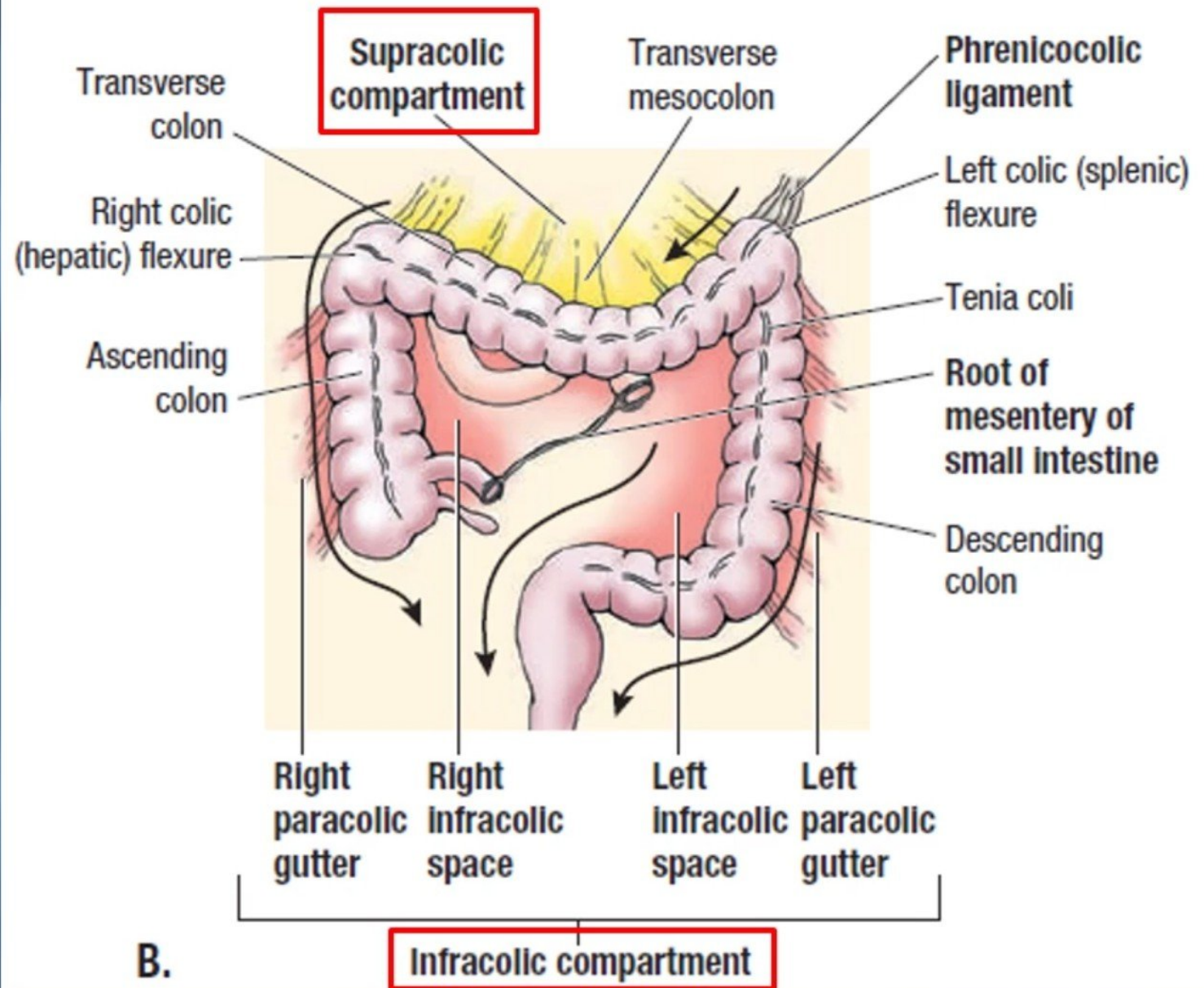
✓ **Infracolic.**

✓ **pelvic.**

- The dividing line between the supracolic and infracolic compartments is the attachment of **the transverse mesocolon** to the posterior abdominal wall, or rather to the organs that lie on the abdominal wall at this level.



✓ Supracolic.
✓ Infracolic.



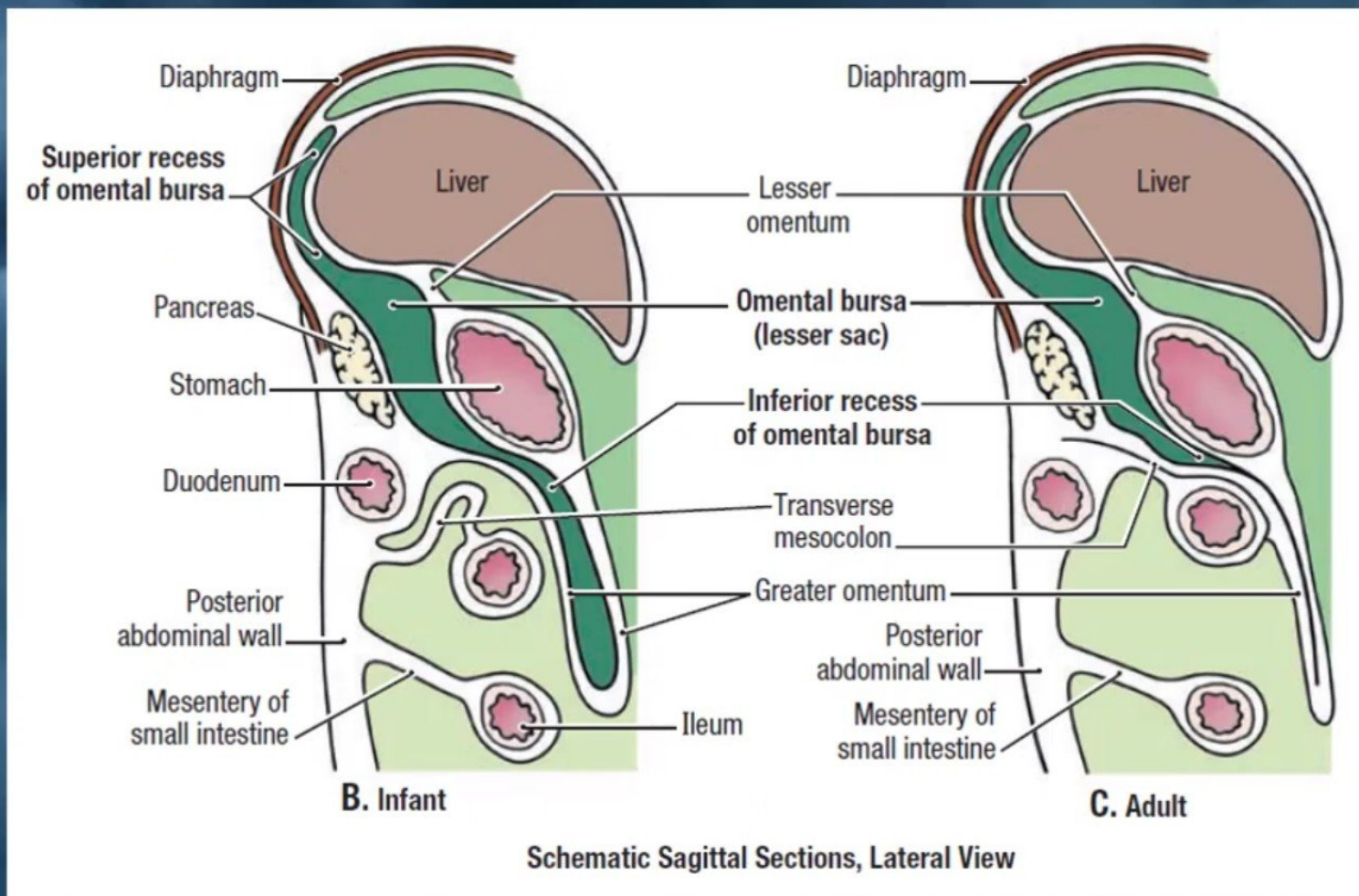
The transverse mesocolon
is a double fold of
peritoneum passing:

- ✓ From **the transverse colon**
- ✓ to - the front of **the second part of the duodenum**,

- the anterior aspect of **the head** and the anterior border of **the body of the pancreas**.

- The transverse colon and transverse mesocolon are adherent to the posterior surface of the greater omentum.

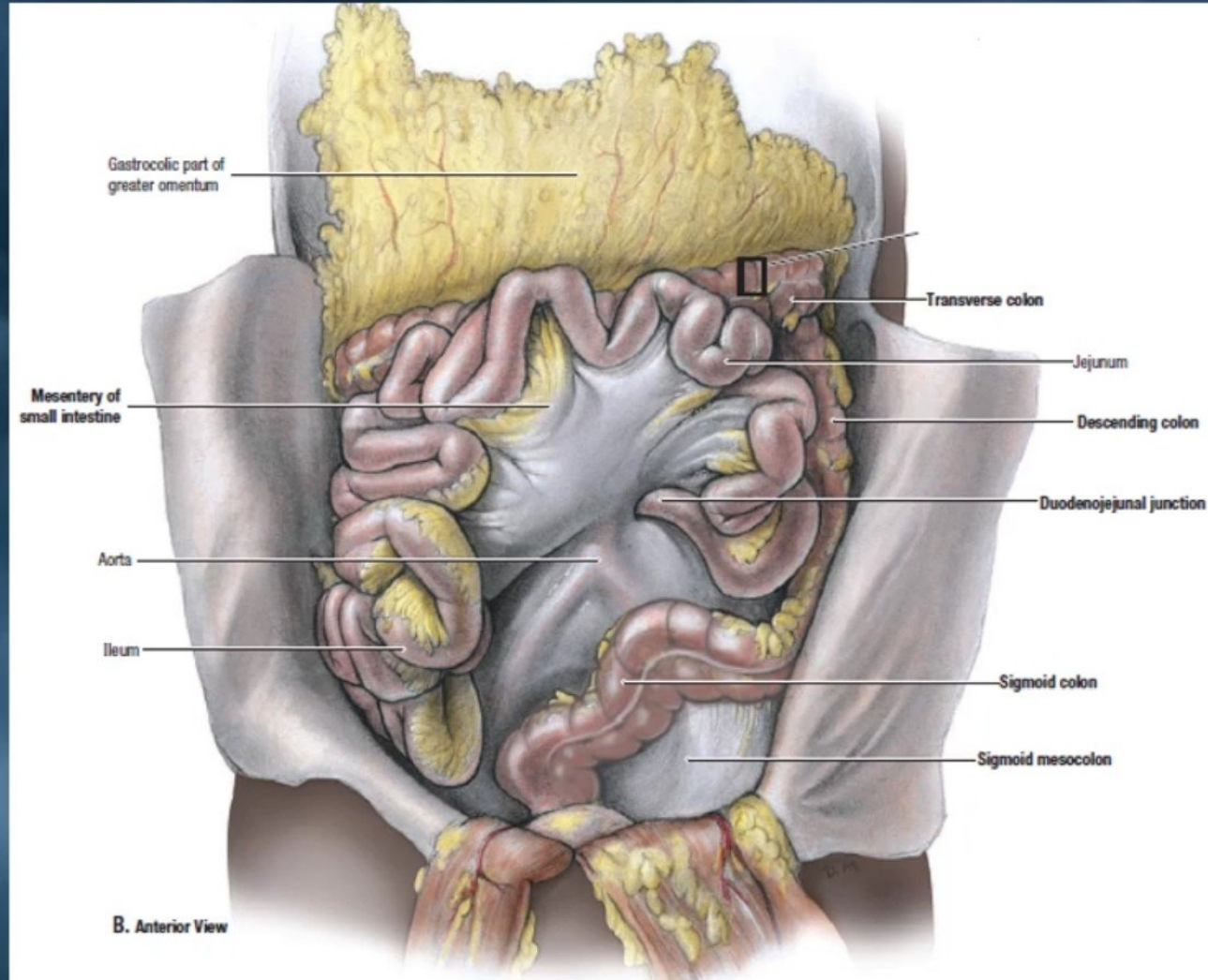
Peritoneal compartments

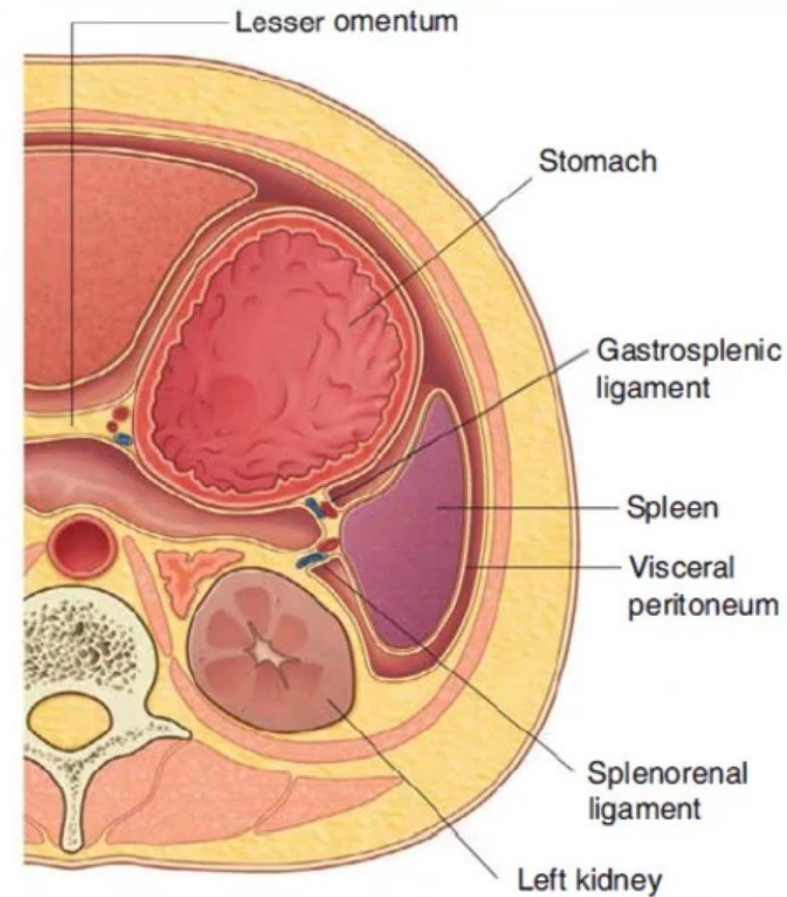
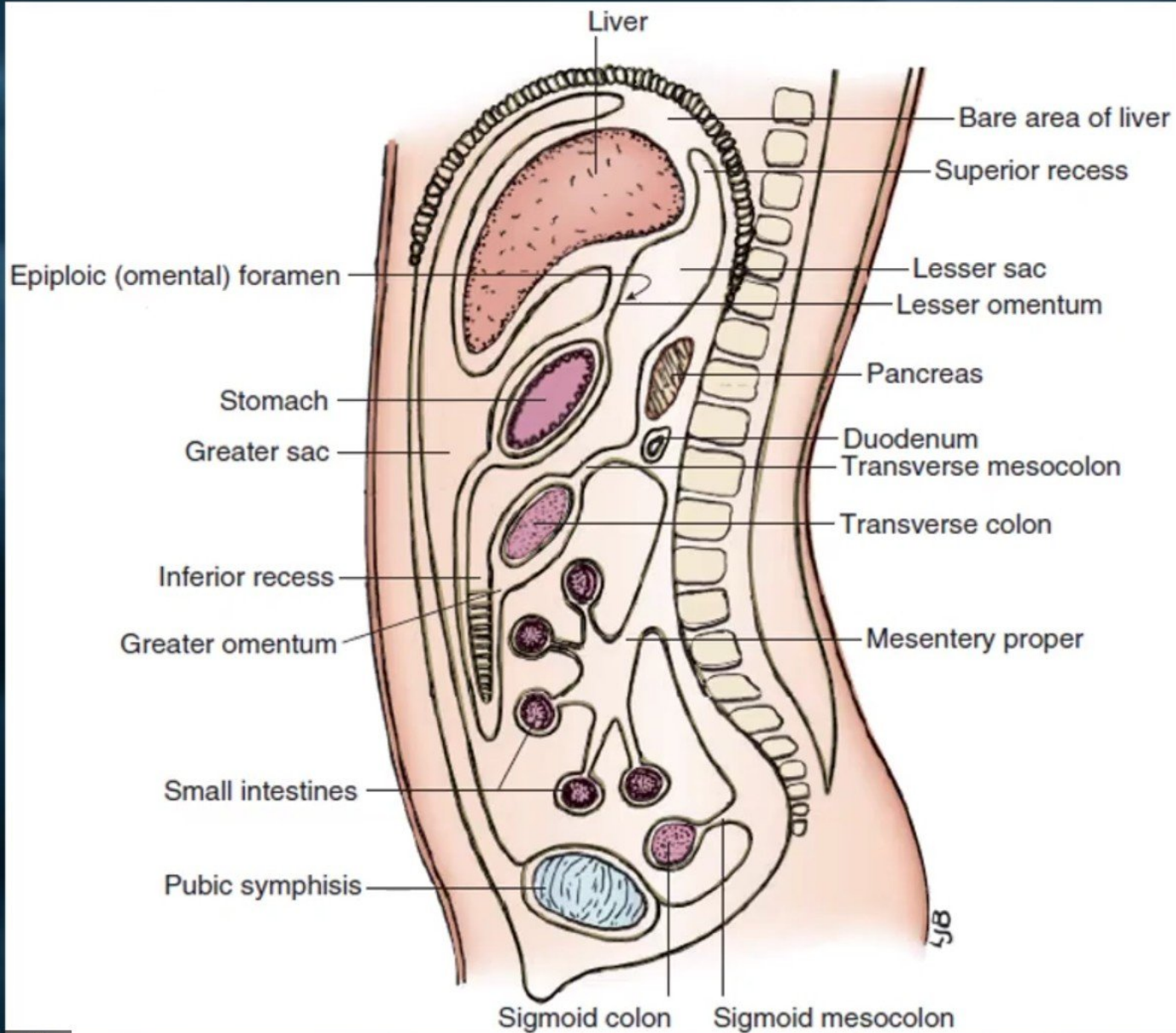


Peritoneal compartments

When you need to see the infracolic compartment:

- When the greater omentum is lifted up over the costal margin, the stomach, transverse colon and mesocolon are lifted upwards with it, and the posterior surface of the mesocolon and **the infracolic compartment** brought into view.





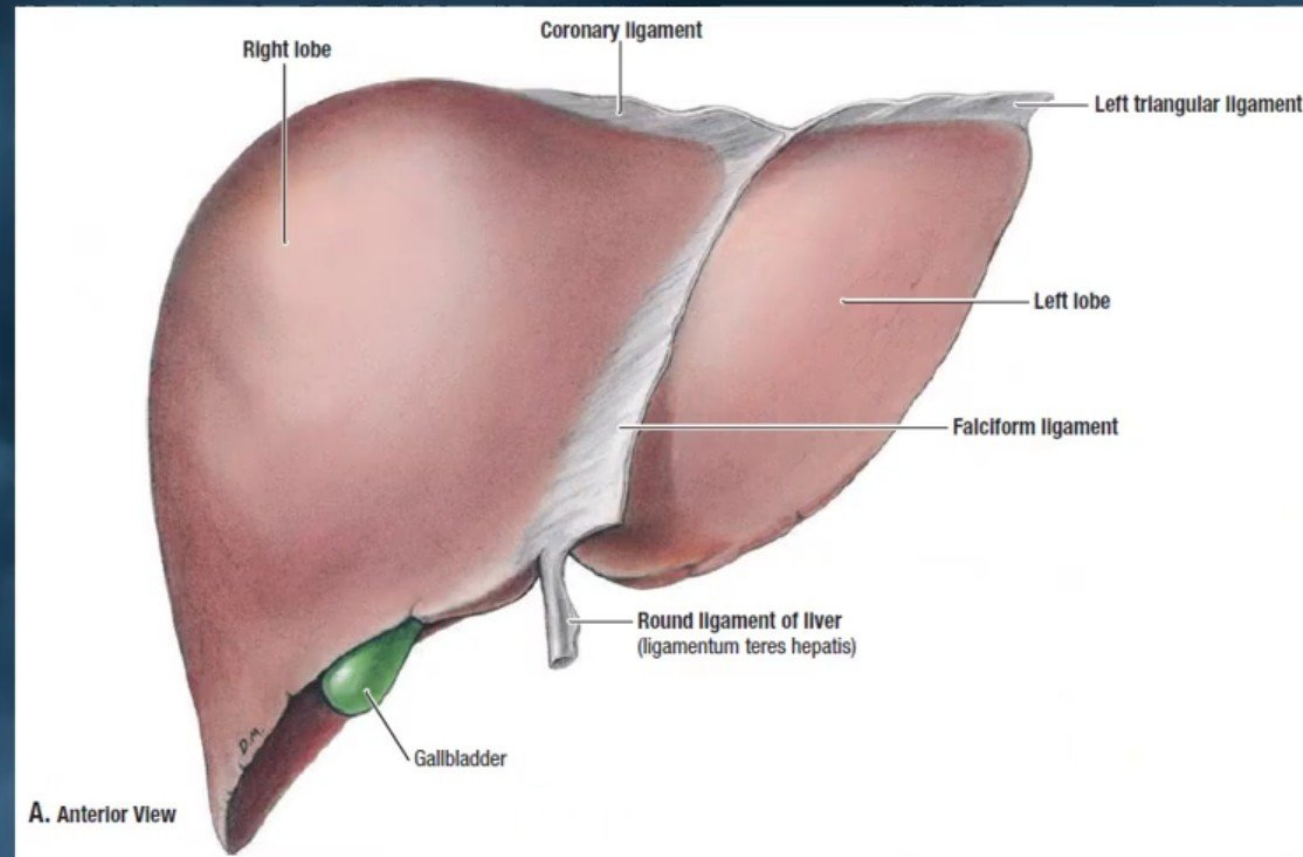
Peritoneal Ligaments

1. Falciform Ligament

- It is connecting **the liver** to **the diaphragm** and **the anterior abdominal wall**.
- It contains the **ligamentum teres hepatis** and the **paraumbilical vein**.

2. Ligamentum Teres Hepatis (Round Ligament of the Liver)

- It lie in the free margin of the falciform ligament and ascends from the umbilicus to the inferior (visceral) surface of the liver,
- It is formed after birth from the remnant of the **left umbilical vein**.



3. Phrenicocolic Ligament

- It is from the **left colic flexure** to **the diaphragm**.

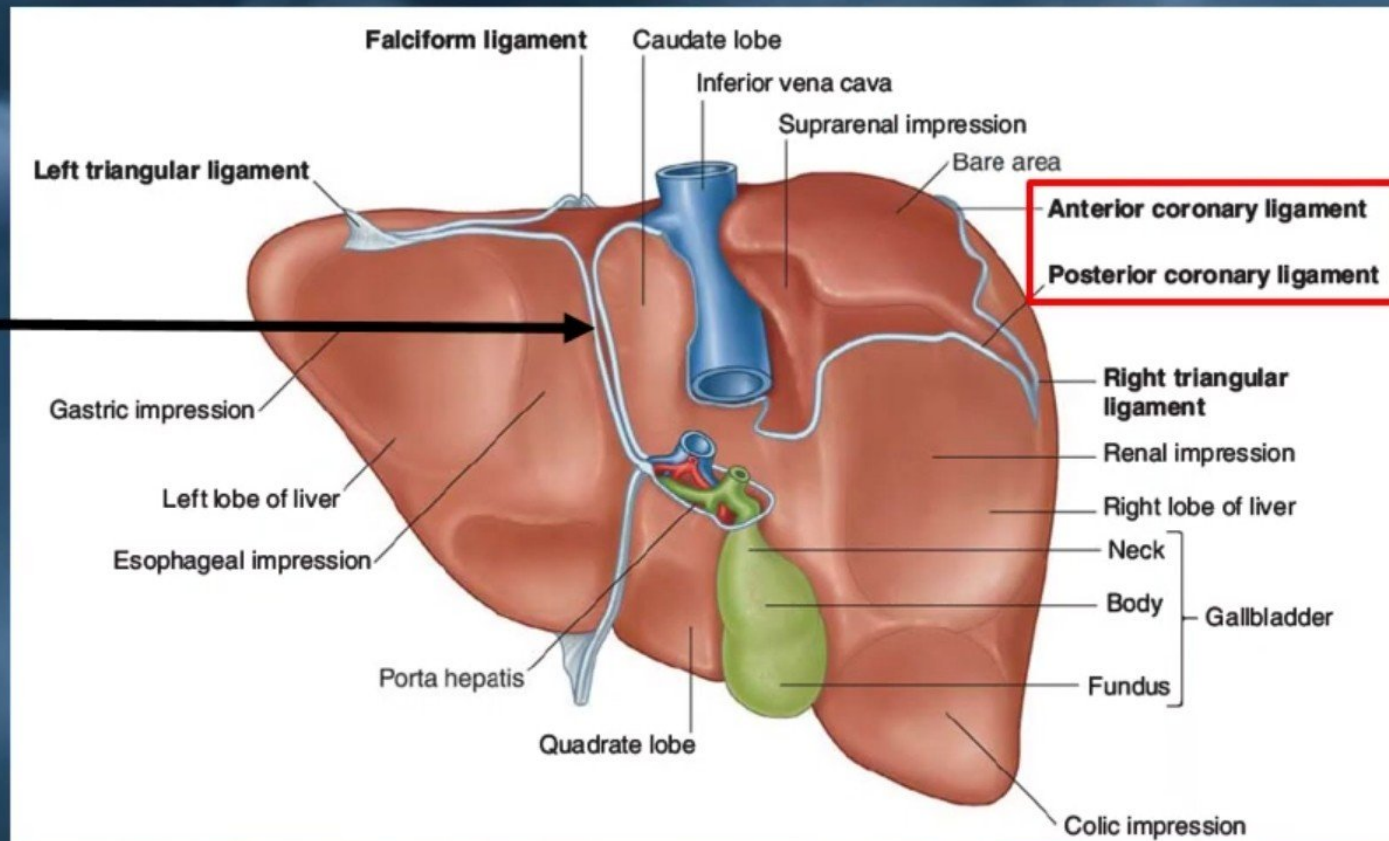
4. Coronary Ligament

- It is a peritoneal reflection from the diaphragmatic surface of the liver .
- It form the **bare area of the liver.**
- It form the **right and left triangular ligaments.**

5. Ligamentum Venosum

- It is the fibrous remnant of the **ductus venosus.**
- It lies in the fissure on the inferior surface of the liver, forming the left boundary of the **caudate lobe of the liver.**

Peritoneal Ligaments



Peritoneal Folds

1. Umbilical Folds

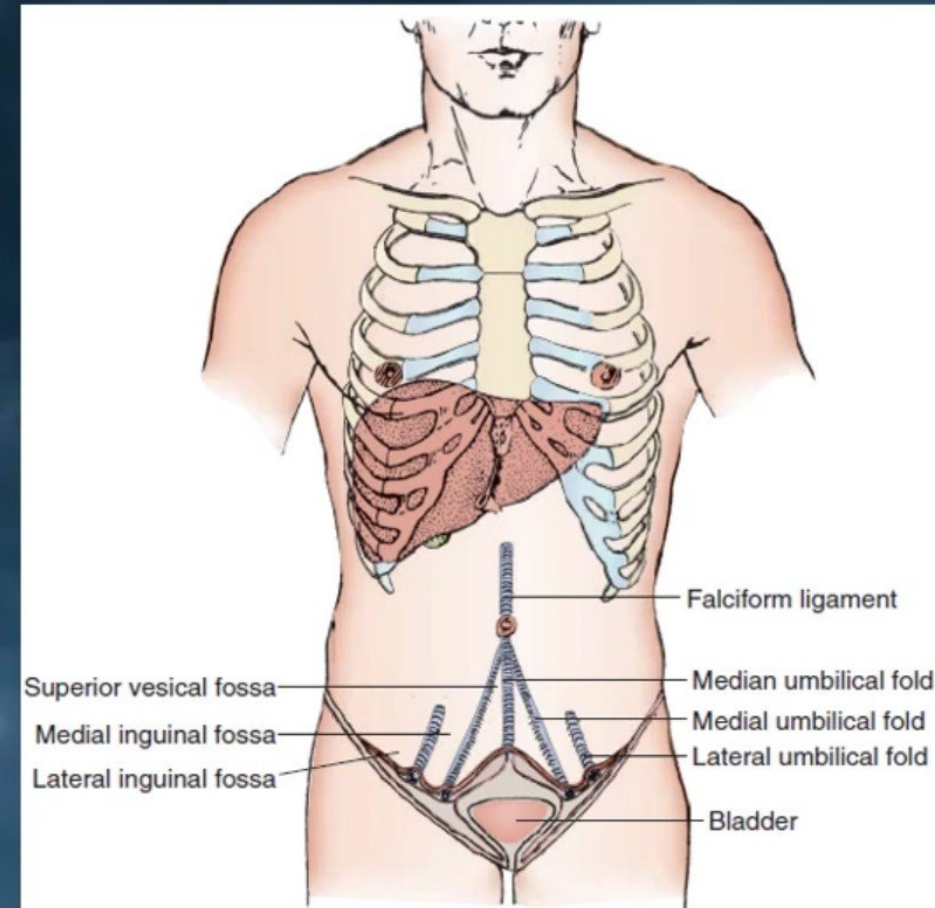
- They are five folds of peritoneum below the umbilicus, including the median, medial, and lateral umbilical folds.

2. Rectouterine Fold

- It extends from the cervix of the uterus,
- It forming the rectouterine pouch (of Douglas).

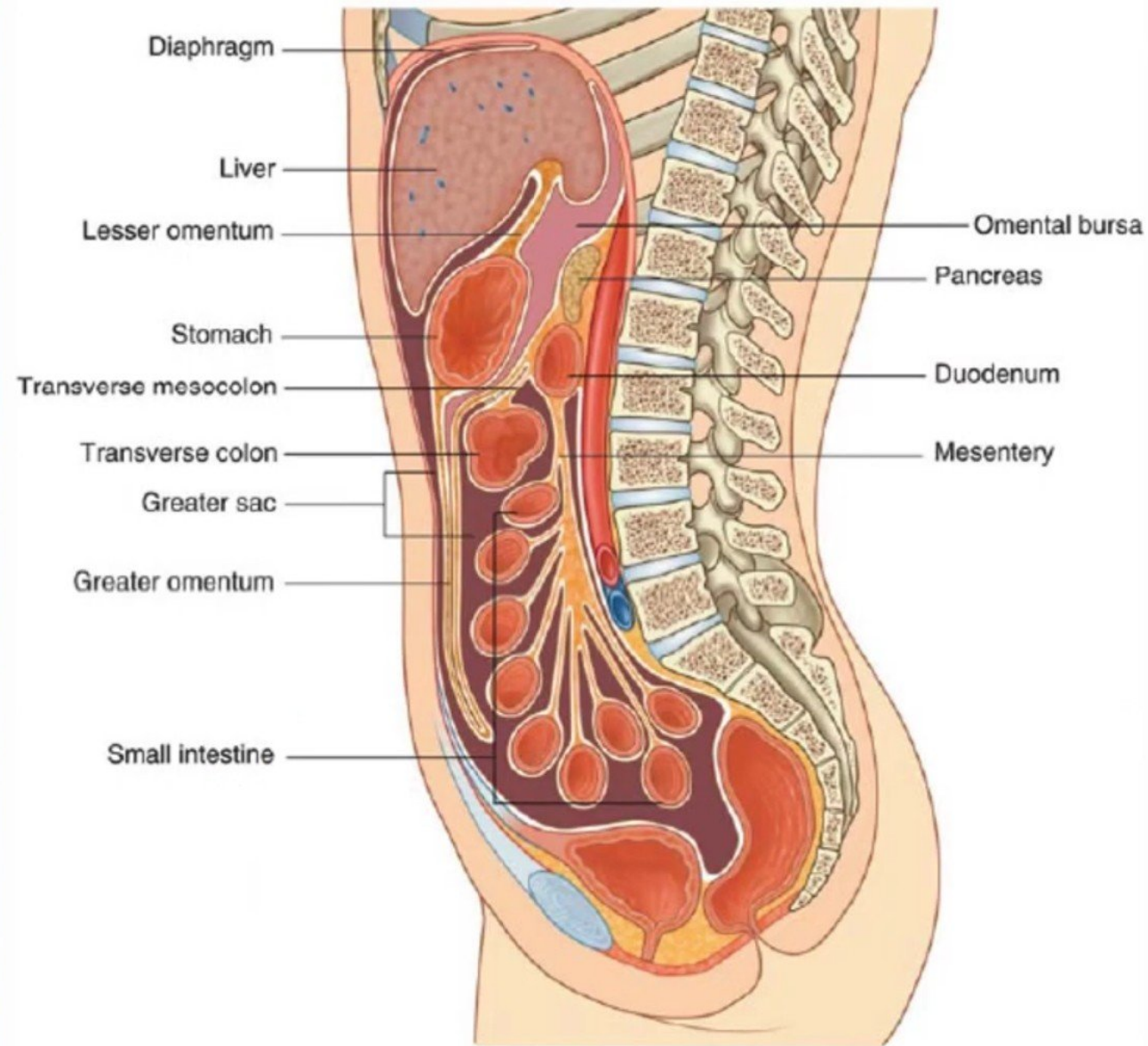
3. Ileocecal Fold

- It extends from the terminal ileum to the cecum.

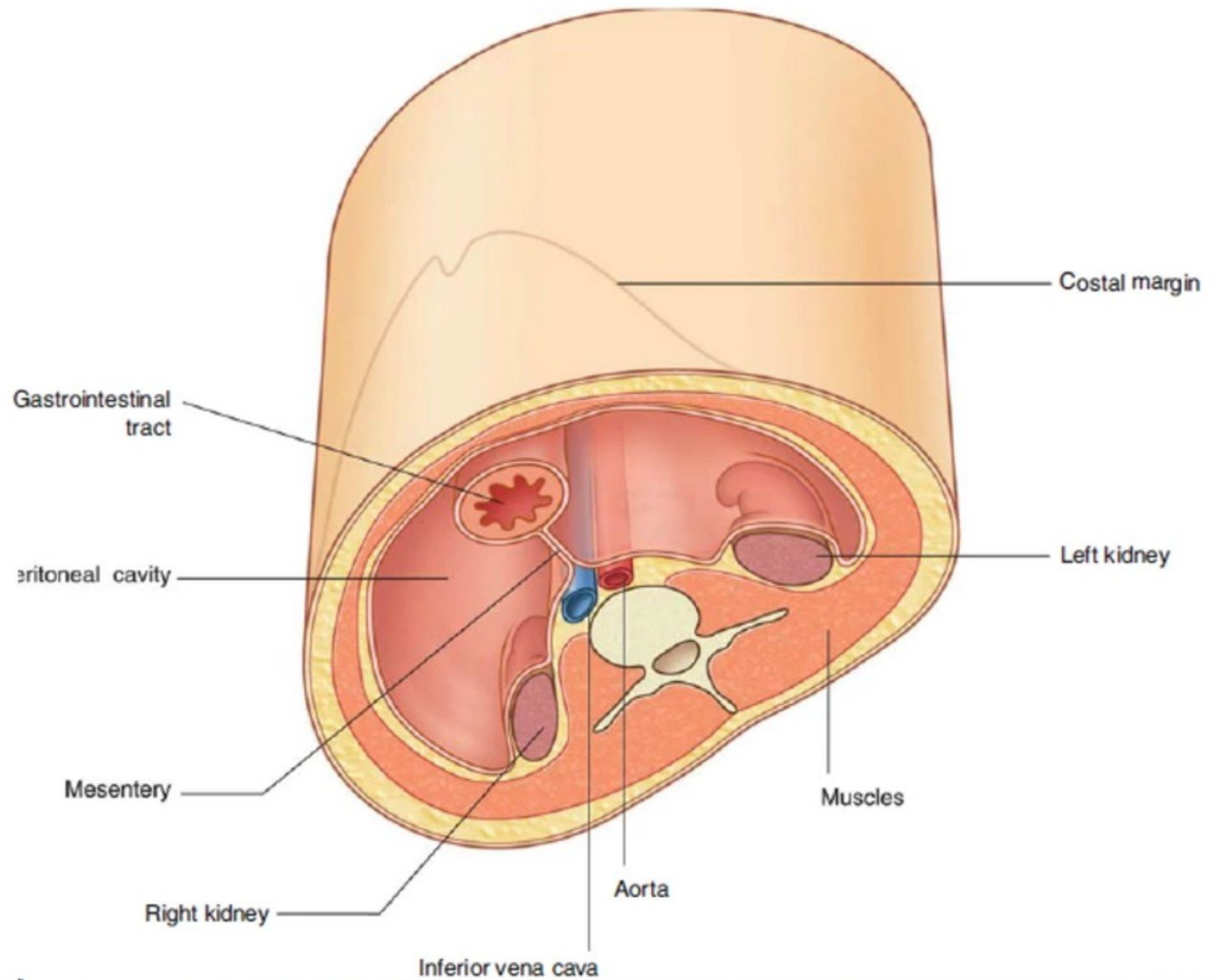
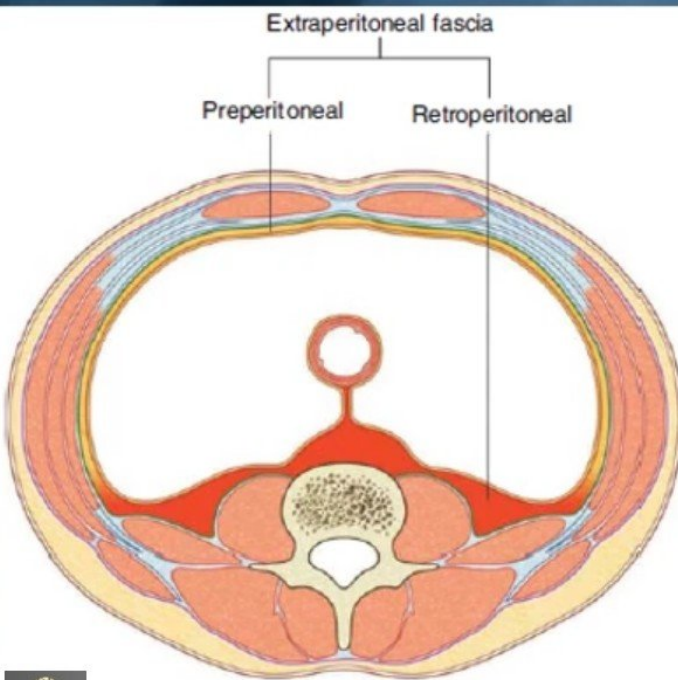


Peritoneal cavity

- It is a **potential space** between the **parietal** and **visceral peritoneum** and
- It contains of **fluids**.
- It is a **completely closed sac** in the **male**.
- It is **open sac** in **the female** through the **uterine tubes**, **uterus**, and **vagina**.
- It is divided into **the lesser** and **greater sacs**.



Peritoneum



Greater Sac

- It extends across the entire breadth of the abdomen and from the **diaphragm** to the **pelvic floor** and **presents numerous recesses**.

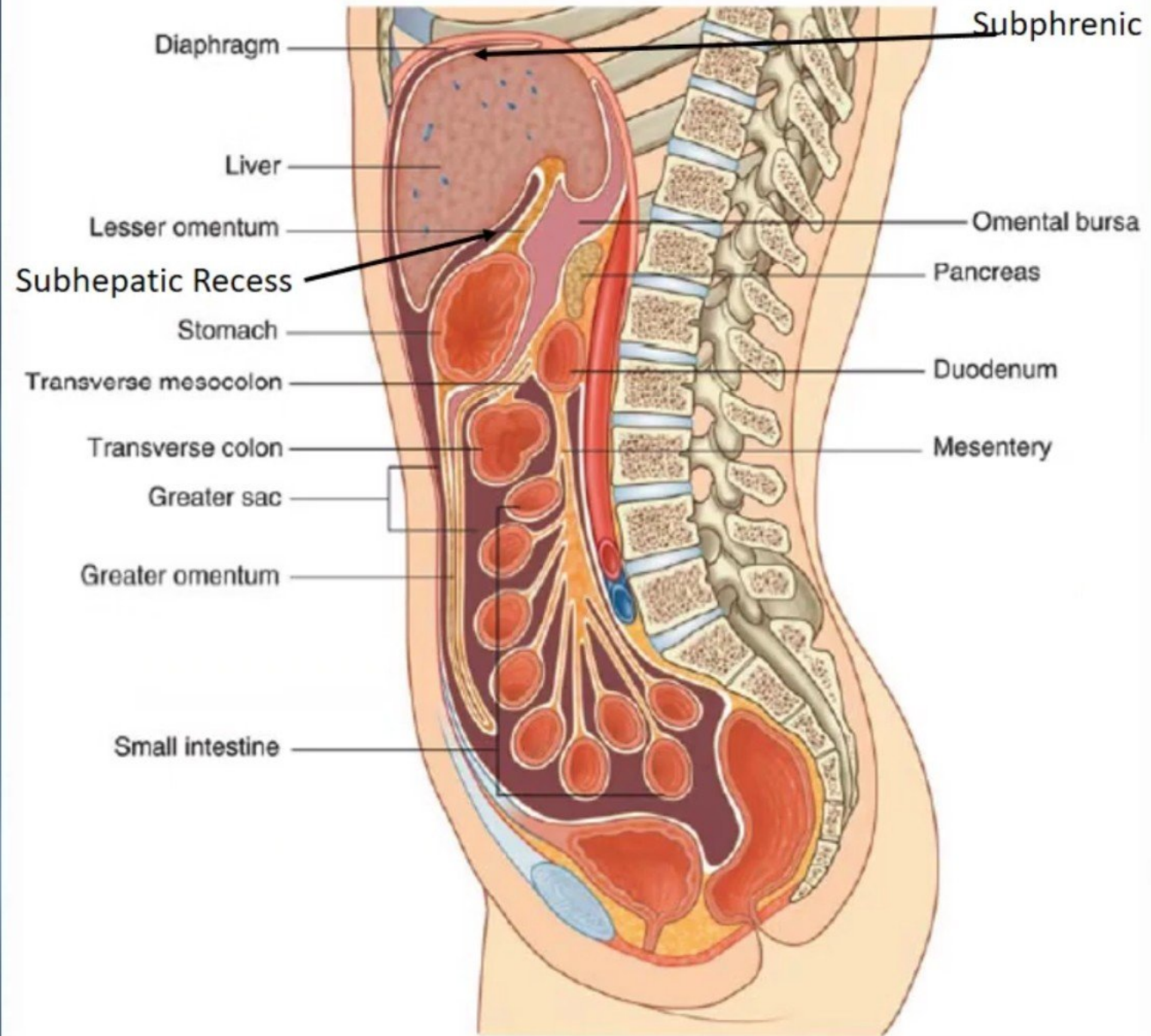
1. Subphrenic (Suprahepatic) Recess.

2. Subhepatic Recess or Hepatorenal Recess (Morrison's Pouch).

- ✓ It is a deep peritoneal pocket between the liver anteriorly and the kidney and suprarenal gland

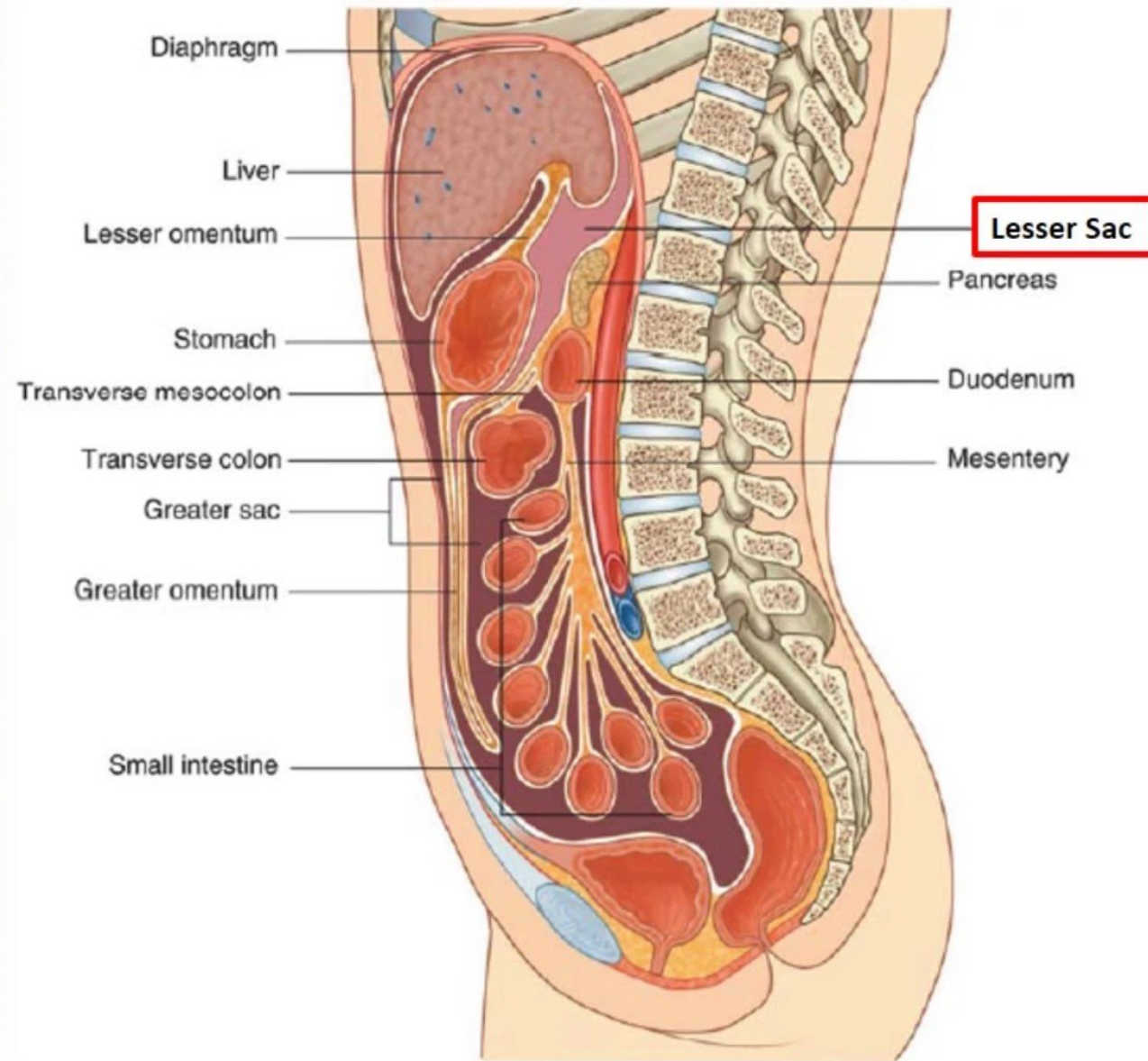
3. Paracolic Recesses (Gutters).

- ✓ It lie lateral to the ascending colon (right paracolic gutter) and lateral to the descending colon (left paracolic gutter).



Lesser Sac (Omental Bursa)

- Its space lies **behind** the **liver, lesser omentum, stomach**, and upper anterior part of the greater omentum.
- It is a **closed sac**.
- its communication with the greater sac through the **epiploic (omental) foramen**.



Thanks